



2026 Regulations

Revision C
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Organized by Innovators Educational Foundation

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SECTION 1 – ADMINISTRATION

1 Purpose

1.1 Fundamental Vision

Collegiate level solar car Events hosted by Innovators Educational Foundation (IEF) seek to promote and celebrate educational excellence and engineering creativity. Fueled by the spirit of friendly competition and teamwork, the Events champion the creative integration of technical and scientific expertise across a range of exciting disciplines.

1.2 Missions

To support and encourage bright young minds to succeed in the technical fields of engineering, the sciences, mathematics, and business, through multi-disciplined experiential learning which in turn enables success in future careers.

To create public awareness and enthusiasm, both for education excellence and engineering creativity itself, and for the technologies and practices that emerge from that excellence.

2 Administration

2.1 The Event

The Event consists of technical inspection of the entries (Scrutineering), the Electrek American Solar Challenge cross-country tour (ASC), and/or the Electrek Formula Sun Grand Prix track competition (FSGP), as described in Reg 6.

FSGP is not in any way associated or affiliated with the Formula 1 companies, FORMULA 1 racing, or the FIA Formula One World Championship.

2.2 Organizers

2.2.A Innovators Educational Foundation

IEF shall be the official Organizers of the Event.

2.2.B International Solarcar Federation

FSGP and ASC are recognized as International Solarcar Federation (ISF) events.

2.3 Event Website

For official information about the Events, registration documents, regulations, participating teams, sponsors, and contact information, visit <https://www.AmericanSolarChallenge.org>. Teams that maintain an internet presence are requested to incorporate a link to the Event Website. The Event webmaster can be reached by email at ASCwebmaster@AmericanSolarChallenge.org.

2.4 Headquarters

Headquarters is responsible for team coordination and Event registration. During the Event, a physical Headquarters will be established at a site appropriate to each Event component and will assume the management functions for the Event. The Headquarters team can be reached by email at ASCteams@AmericanSolarChallenge.org.

2.5 Officials

A team of Officials will be selected by the Organizers to conduct the Event. Officials having specific duties shall be announced to the teams through Event briefings.

2.5.A Event Director

The Organizers will appoint an Event Director who has responsibility for the Event and can be reached by email at ASCinfo@AmericanSolarChallenge.org.

2.5.B Regulations Manager

The Organizers will appoint a Regulations Manager who is responsible for the Event Regulations and can be reached by email at ASCregs@AmericanSolarChallenge.org. The Regulations Manager reports to the Event Director and Organizers.

2.5.C Chief Inspector / Clerk of Course

The Organizers will appoint a Chief Inspector / Clerk of Course who is responsible for solar car Scrutineering inspections and enforcement of the Regulations during the Event. The Chief Inspector / Clerk of Course reports to the Event Director and Organizers.

2.5.D Head Timekeeper

The Organizers will appoint a Head Timekeeper who manages Timing Officials and is responsible for Event timing and scoring. The Head Timekeeper reports to the Chief Inspector and Organizers.

2.5.E Technical, Safety and Fairness Officials

The Event Director, Chief Inspector, Regulations Manager and Head Timekeeper will appoint additional Officials to help conduct the Event.

2.5.E.1 Inspectors: A team of “Inspectors” will be assigned to grade design reports, perform Scrutineering inspections, and enforce the Regulations.

2.5.E.2 Staff: A team of “Staff” will support the Event and will be Officials during the Event.

2.5.E.3 Track Steward: During FSGP, an Official will be assigned as the “Track Steward”. The Track Steward will have authority and control over operations of the track.

2.5.E.4 Observers: During ASC, a team of “Observers” will be assigned to monitor the progress of the teams and report back to the Inspectors.

2.5.F Jury

A Jury will be available to teams during the Event which can evaluate protests on conformity with these Regulations, resolve team disputes, and rule on penalty appeals. In addition, the Jury is empowered to decide cases not specifically covered by these Regulations. The Jury will consist of a minimum of three (3) distinguished individuals who have experience with solar car events.

2.6 Application of Regulations

These Regulations will apply to all aspects of the Event including team registration, Scrutineering, FSGP, and ASC.

2.7 Acceptance of Regulations

All persons or groups selected to participate in the Event are assumed to know these Regulations. Their participation in the Event will constitute acceptance of them.

2.8 Interpretation of Regulations

Only the Regulations Manager, Chief Inspector/Clerk of Course, and Inspectors are authorized to interpret the Regulations.

2.8.A Official / Unofficial Interpretations

2.8.A.1 Teams shall identify if they are requesting an Official or Unofficial Interpretation of the Regulations.

2.8.A.2 Official Interpretations will have the same force and effect as the Regulations and will be responded to such that all teams have visibility to the question and response.

2.8.A.3 Unofficial Interpretations will have no force and effect on the Regulations, may be superseded, and will be kept private between the team and the Inspectors.

2.8.B Prior to Scrutineering

2.8.B.1 Teams requesting interpretation of the Regulations shall submit their question(s) to the Inspectors through email at ASCregs@AmericanSolarChallenge.org.

2.8.B.2 All Official Interpretations will be posted to the regulations page on the Event Website.

2.8.C During and after Scrutineering

2.8.C.1 All Official Interpretations will be announced at Briefings as well as on the Event Website.

2.9 Supplemental Documents

2.9.A Additional Documents

Additional documents may be distributed to all teams entered in the Event to supplement these Regulations. These documents will clearly state that they are a supplement to the Regulations and they will have the same force and effect as these Regulations.

2.9.B Document Conflict

If there is a conflict between a supplemental document and these Regulations, the document having the later date shall take precedence. The Organizers reserve the right to revise these Regulations at any time.

2.10 Advertising, Promotion, and Publicity

All advertising, sales promotion, and publicity material produced by the teams or their sponsors concerning or referring to the Event will refer prominently to the Event as “The Electrek American Solar Challenge” and “The Electrek Formula Sun Grand Prix”. All teams, by entering the Event, specifically agree to abide by this regulation. Furthermore by entering the Event, all teams and team members agree to the use of their names and their likenesses in any publicity materials (brochures, magazines, videos, photographs, etc.) that may be issued by the Event’s sponsors or Organizers.

2.11 Event Cancellation

The Organizers reserve the right to cancel, abandon, or postpone the event, in whole or in part, for any reason, including but not limited to acts of God, natural disasters, acts of war, acts of terrorism, weather, riots, strikes, communicable disease crisis, quarantine restrictions, and other events beyond the control of the Organizers.

3 Safety

Each team is responsible for the road-worthiness of its solar car. Passing Event components of Scrutineering and FSGP or implementing changes suggested in comments on the team's design reports does not relieve the team of any liability.

All solar cars and support vehicles must be maintained in a safe, road-worthy condition and be operated safely and within the law at all times. A team may be disqualified and withdrawn from the Event at any time if it is judged to be operating in an unsafe manner. All solar cars and support vehicles are operated and driven at the team's own risk.

Each team is responsible for the safety of its members, and any minimum criteria specified by the Organizers via these Regulations and/or correspondence between the teams and the Organizers should not be construed as design specifications for the construction of a "safe" solar car.

Teams must look after the health and safety of their team members.

3.1 Team Safety

3.1.A Team Safety Officer

- 3.1.A.1** Each team is required to have at least one Safety Officer.
- 3.1.A.2** All Safety Officers shall be trained in basic First Aid including CPR. Proof of training shall be submitted to Headquarters with the Team Roster (available on the Event Website).
- 3.1.A.3** At all times, one of the team's Safety Officers must be the Designated Safety Officer. The Designated Safety Officer must not hold the role of Solar Car Driver, Solar Car Passenger, Support Vehicle Driver, or Project Manager while acting as the Designated Safety Officer. The Designated Safety Officer must travel in the Lead or Chase Vehicles while the Solar Car is on the road, and be present in the pit area while the Solar Car is on the track.

It is encouraged to have more than one team member who is trained in basic First Aid including CPR.

3.1.B Team Safety Kit

- 3.1.B.1** Each team shall have available the following safety kit:
 - 1) First-aid kit meeting ANSI Z308.1 Class A or B, Type III or IV;
 - 2) Safety vests (1 per occupant);
 - 3) Four (4) orange cones (minimum 300 mm high);
 - 4) Two (2) orange warning flags;
 - 5) ABC fire extinguishers (30 lb / 13.5 kg¹ active material capacity or larger);
 - 6) Battery MSDS;
 - 7) Battery spill kit / method of containment of battery fires / 40 kg of sand;
 - 8) A shovel/spade (for applying the sand);
 - 9) Safety glasses and gloves for handling batteries; and,
 - 10) A 5 gallon metal bucket for inerting and disposal of damaged electrochemical cells.
- 3.1.B.2** The above safety kit shall be distributed among the applicable support vehicles during ASC, and shall be located at the pit while the solar car is on the track for FSGP. Items 5 through 10 shall travel with the solar car between Scrutineering stations.
- 3.1.B.3** The team shall ensure that applicable Personal Protective Equipment (PPE) is used for fabrication, repairs etc., i.e. safety glasses, cut resistant gloves etc.

¹ The 30 lb / 13.5 kg fire extinguisher requirement should be achieved with a set of extinguishers that can meet Regs 12.5.B.4 and 12.5.C.3. Example extinguishers that would meet the requirements could be one 20 lb and one 10 lb extinguisher, or three 10 lb extinguishers.

4 Entries

4.1 Event Registration

The Event is only open to institutions of higher education (participants). No participating team will be considered fully registered until all items below are received by Headquarters and marked Green on the Team Status Board on the Event Website.

4.1.A Registration – Initial Registration Package

4.1.A.1 Each team wishing to participate must submit an Initial Registration Package consisting of:

- 1) Team Entry Form;
- 2) Team Participation Agreement;
- 3) Proof of Insurance;
- 4) Preliminary Vehicle Design Report (as described in Reg 5.2); and,
- 5) The initial entry fee, US\$1900.

4.1.A.2 Other Event registration items will not be processed before receiving the Team Entry Form.

4.1.A.3 This portion of the entry fee is non-refundable. Inspectors are not obligated to grade design reports for teams who have not yet paid the Initial Entry Fee and they will prioritize teams who have paid this fee.

4.1.B Registration – Track Registration Package

4.1.B.1 Each team must submit a Track Registration Package consisting of:

- 1) Vehicle Design Report (as described in Reg 5.3); and,
- 2) The track fee, US\$4150.

4.1.B.2 All teams planning to participate in ASC must also participate in FSGP.

4.1.B.3 This portion of the entry fee is non-refundable.

4.1.C Registration – Road Registration Package

4.1.C.1 Teams planning to compete in ASC must also pay a road fee of US\$3900.

4.1.C.2 Teams desiring to only participate in FSGP do not need to pay this fee.

4.1.C.3 This fee is partially refundable if the team withdraws from the Event. US\$2300 will be refunded if written notification is submitted to and confirmed by Headquarters no later than the date listed in Reg 4.2.A.4 No refunds will be made after that date.

4.2 Registration Deadlines & Payment

4.2.A Registration Dates

4.2.A.1 Initial Registration Package is due: December 1, 2025

4.2.A.2 Track Registration Package is due: January 31, 2026

4.2.A.3 Road Registration Package is due: March 15, 2026

4.2.A.4 Refund cut-off is: May 15, 2026

4.2.A.5 Vehicle Data Sheet, Photo, and Team Roster are due: May 1, 2026

4.2.A.6 Public Release of Vehicle Data Sheets and Photos: May 15, 2026

4.2.A.7 Build Review is due: June 17, 2026

4.2.B Late Penalties

Late penalties will be charged for fees and reports received more than ten (10) days past the deadline.

4.2.B.1 A US\$200 penalty will be charged for late submission of the Initial Registration Package.

4.2.B.2 A US\$400 penalty will be charged for late submission of the Track Registration Package.

4.2.B.3 A US\$600 penalty will be charged for late submission of the Road Registration Package.

4.2.C Entry Fee Payments

- 4.2.C.1 Transaction Fees:** Any additional fees resulting from payments made to IEF for entry into the Event shall be the responsibility of the team making payment utilizing such methods incurring the added fees.
- 4.2.C.2 Foreign Currency:** It is the responsibility of the team for making payment in US currency. IEF is not obliged to accept payments made in any currency other than US dollars.
- 4.2.C.3 Payment Via ACH:** If your school has the ability to make electronic ACH payments, this can be a good option for quick payments without transaction fees. To start the process of adding IEF as a vendor for ACH payments, have your school's procurement person contact IEF@AmericanSolarChallenge.org.
- 4.2.C.4 Payment Via Check:** Make checks payable to Innovators Educational Foundation and mail them to the IEF address listed on the Event Website.
- 4.2.C.5 Payment Via PayPal:** Teams can pay with a credit/debit card or bank account using PayPal. This results in transaction fees of US\$0.49 per payment plus 1.99% of the transaction (or 3.49% for payments from outside the United States). To use this method, first calculate the PayPal Transaction Amount using the following formula and then pay this amount to IEF@AmericanSolarChallenge.org in USD.
- Domestic US PayPal Payment Amount = (Payment Due + 0.49) / 0.9801
International PayPal Payment Amount = (Payment Due + 0.49) / 0.9651
- 4.2.C.6 Payment Via Wire Transfer:** A US\$50 wire transfer transaction fee shall apply to all wire transfers, which covers IEF bank fees for accepting incoming wire transfers. This fee is in addition to any transaction fees charged by the issuing bank and should be included with the payment. Contact IEF@AmericanSolarChallenge.org if your team is interested in paying via Wire Transfer.
- 4.2.C.7** Payments made will be applied against the registration packages in order, i.e. payment will be applied to the Initial Registration Package first, followed by the Track Registration Package etc.

4.3 Grants

Teams with demonstrated financial need, and technical potential, may apply for a grant from the Event Organizers to cover a portion of their entry fees. Teams will be required to submit a proposal detailing their request for Grant funding. The proposal form will be provided to the team by Headquarters upon request. Award of a Grant is at the discretion of the Organizers.

4.4 Team Data

Each team must submit a team photo, vehicle data sheet, and team roster to Headquarters by the date in Reg 4.2.A.5. The photo and vehicle data can be publicly released and used in Event brochures, on the event website, etc. Late submissions may be omitted. Early submissions will not be made public prior to the date in Reg 4.2.A.6 without permission of the team representative. After this date, team information can be released by Headquarters. Teams must submit forms for each of the following in the format provided on the website

4.4.A Team Photo

Each team shall provide a digital team photo that clearly shows the solar car and team members. The photo must be landscape, 3:2 aspect ratio, at least 1080x720 pixels, and in jpg or png format. In addition, teams must submit a filled-out Team Photo Form, which can be accessed via the website and contains a team bio along with name identification of the team members pictured.

4.4.B Vehicle Data Sheet

Each team shall provide a data sheet which must include solar car name, weight (road-ready, without occupants), dimensions, motor configuration and rating, solar cell type and manufacturer, estimated solar collector power at STC per Reg 5.3.E.6, battery energy capacity from the Battery Approval Form (Appendix G), battery type and manufacturer, chassis description, and wheel specifications and count, tire manufacturer and size. MOV teams additionally must provide occupancy, cargo capacity, and charging information. All specifications must be provided in the specified metric units. .

4.4.C Team Roster

Each team shall provide a Team Roster which must list all team members and designate the team leader, safety officer(s), solar car drivers/passengers, and faculty advisor(s). A copy of the front of each solar car driver's valid Drivers License must be provided with the Team Roster. See Reg 3.1.A for Safety Officer requirements. Proof of First Aid/CPR training must be submitted with the Team Roster. Teams needing to make changes to their roster after the initial submission must submit an updated roster to Headquarters electronically.²

4.5 Number of Team Members

Entry fees cover up to fifteen (15) team members.³ Beyond this, a US\$80 fee per extra team member applies.

4.5.A Supporters and Additional Team Members

The Organizers reserve the right to determine if any individual is acting de facto as part of a team.

4.6 University/College Advisor

4.6.A.1 All teams must have at least one University / College advisor ("Advisor") who will provide guidance as needed throughout the solar car design, building, and testing process.

4.6.A.2 The University / College advisor shall be an employee of the team's host institution be it a University or College etc., that is the official acting on behalf of the educational institution representing the participating entry into the Event.

4.6.A.3 It is strongly recommended that the Advisor attend the Event.

4.7 Individual Participant Registration

All individuals participating in the Event must be registered with Headquarters, including team members, sponsors, officials, guests, and media personnel. All participants must present themselves at Registration to complete all required forms. Badges will be issued and used to obtain access to restricted areas. These badges must be visible at all times.

4.8 Event Correspondence

4.8.A Primary Team Contacts

Communication with Event Officials shall be conducted in English, the official language of the Event, by team leaders (Faculty Advisor, Project Manager, & Department Leads) as detailed on the Team Entry Form.

4.8.B Emails

4.8.B.1 Email is the preferred communication method for Event correspondence. Participating institutions must set up a Generic Team Email account for this purpose that forwards messages to the current team leaders (Faculty Advisor, Project Manager, & Department Leads). The team is responsible for ensuring that as team leaders are replaced the email forwarding is updated accordingly.

4.8.B.2 The Generic Team Email address must be included on all email correspondence between any team member and any Event representative.

4.9 Insurance

All teams need to maintain vehicular liability and general public liability insurance with limits of liability for (1) bodily injury of not less than US\$1,000,000.00 for each person and US\$1,000,000.00 for each occurrence, and for (2) property damage of not less than US\$1,000,000.00 for each accident and US\$1,000,000.00 in the aggregate. Teams will be required to provide a certificate of such insurance or proof of self-insurance.

² Teams are encouraged to submit necessary changes to the team roster as early as possible. Teams making changes to their team roster less than 2 weeks prior to the start of the event or at the event can expect to be delayed in on-site registration and check-in.

³ It is understood that team members may arrive and depart from the Event at various times. The number of team members listed in the regulation is the total number registered, not the number of team members present at any one time.

4.10 Withdrawals and Exclusion

4.10.A Voluntary

The team's Faculty Advisor or Project Manager must notify Headquarters in writing.

4.10.B Not Meeting Requirements

Headquarters may withdraw teams that do not meet the registration deadlines or fail to present a solar car that satisfies Scrutineering or Event requirements.

4.10.C Exclusion

Exclusion will occur if the Officials deem a team to have departed from the spirit of the Event by deliberately acting to gain unfair advantage over other teams or to have acted in an unsafe manner.

4.11 Donations

Teams that withdraw after making payment will have funds considered as donations to IEF in consideration that no services or goods are provided for said funds.

4.12 Team Numbers

Each team registered for the Event will have a unique number approved by Headquarters (positive integer, three (3) digits maximum).

4.12.A.1 Number Retention / Reservation: Teams which have participated in four (4) or more previous ASC (Sunrayce, NASC) cross-country events have the right to reserve their team number from those openly available for selection by other teams. The number being selected for reservation must be the number used in the four (4) events.

4.12.A.2 Requests to reserve a number shall be made in writing to Headquarters, with a listing of the events which the team attended and team number.

4.12.A.3 Team numbers (either use of a reserved number or new unallocated number) will be confirmed as teams complete registration paperwork and submit entry fees.

4.12.A.4 Number Conflict: If a conflict in team numbering arises, Headquarters will determine the numbers assigned. Resolution will be based on order of requests and payment of entry fees with respect to when the team number request is made. If a team fails to pay their entry fees on schedule their requested number can become available to another team who is current on their payments.

4.12.A.5 Tracking of Reserved Numbers: Headquarters will track the Reserved Numbers, and will post a list of the Reserved Numbers and the corresponding team on the Event Website.

4.12.A.6 Release of Reserved Numbers: Should a team with a reserved number desire to change their number, the previously reserved number will become unallocated and available to other teams. A team cannot have more than one reserved number.

5 Technical Documentation

5.1 Document Format

Submit separate PDF documents for each design report detailed below (5.2.A, 5.2.B, 5.3.A, 5.3.B, 5.3.C, 5.3.D, 5.3.E). Early submissions may be sent for review as individual reports without the complete package. File names shall be formatted as:

HostInstitutionName_ASC2026_ReportType_YYYY_MM_DD

For example: a mechanical PVDR report from the University of Acme would appear as follows:

“UnivOfAcme_ASC2026_Mechanical_PVDR_2025_12_01.pdf” with the first letter of each new word capitalized and common abbreviations accepted.

5.2 Preliminary Vehicle Design Report (PVDR)

A preliminary vehicle design report package containing an overview of the initial design of the car's mechanical and electrical systems must be submitted to Headquarters for approval. Both PVDR reports detailed below must be submitted by the date in Reg 4.2.A.1.

5.2.A Mechanical PVDR

The mechanical PVDR report shall provide the preliminary mechanical design information on the structural chassis and roll cage protecting the driver. The report must:

5.2.A.1 Include a completely filled out Mechanical PVDR Form (Appendix D) as the cover page and table of contents at the beginning of the report.

5.2.A.2 Consider road and traffic conditions when determining mechanical loading criteria and potential failure modes. For suspension and steering systems, analysis shall include a minimum of 1G turn, a 2G bump, and 1G braking case loads with a worst-case condition of the loads combined including steady-state load transfer. These loads shall be applied at the tire contact patch. Loads shall be provided in dimensional units, along with any relevant assumptions used in the analysis.

5.2.A.3 Describe roll over and impact protection for the driver.

5.2.A.4 Provide windshield material selection and data sheet.

5.2.B Electrical PVDR

The electrical PVDR report shall provide the preliminary electrical and battery system design information. The report must:

5.2.B.1 Provide school name and team number.

5.2.B.2 Attach a preliminary Battery Approval Form from Appendix G for the main and supplemental pack.

5.2.B.3 Provide an overview of the electrical approach to the solar vehicle.

5.2.B.4 Provide a preliminary block diagram of the electrical system showing all wiring greater than 24V.

5.2.B.5 Describe battery types and suppliers/manufacturers' being considered if one is not already chosen.

5.2.B.6 Describe battery testing and characterization plans, as well as methods for selecting matched cells for module construction, with critical dates indicating when verification activities will be performed with respect to module and pack assembly milestones.

5.3 Vehicle Design Report (VDR)

A vehicle design report package detailing the technical design of the solar car's mechanical systems, electrical systems, batteries, and solar collector must be submitted to Headquarters for approval. All five VDR reports detailed below must be submitted by the date in Reg 4.2.A.2. The information contained in each team's final VDR submission must match the solar car presented at Scrutineering. Safety should be the primary concern with regard to the structural development and fabrication of the solar cars.

5.3.A Mechanical VDR

The mechanical VDR report shall present the as-built design and must:

5.3.A.1 Include a completely filled out Mechanical VDR Form (Appendix F).

5.3.A.2 Review and follow the detailed instructions provided in Appendix E.

5.3.A.3 Attach a completed design review form from the Team Status Board.

5.3.B Electrical System VDR

The electrical system VDR shall document the electrical design approach and must:

- 5.3.B.1** Provide school name and team number.
- 5.3.B.2** Attach a Battery Approval Form from Appendix G for the main and supplemental pack.
- 5.3.B.3** Provide a functional system diagram and rough schematic showing all essential power circuits and electrical equipment, including the solar collector. The diagram shall include power generation devices (solar collector, regenerative braking, etc.), energy storage (batteries, capacitors, etc.), Main Power Contactors/switching, battery protection systems, motor, motor controller, Battery Fuse and any auxiliary circuits.
- 5.3.B.4** Provide system power requirements, high power wiring specification, and fuse specification data sheet to support fuse & wiring selection.
- 5.3.B.5** Contain anything modified or not previously presented from the Reg 5.2.B PVDR.

Include applicable impound & MOV information as a separate section of the report:

- 5.3.B.6** Specify Impound in Vehicle or Impound Box.
- 5.3.B.7** Explain how the battery enclosure and all high voltage connections will be locked/sealed within the vehicle (teams planning to Impound in Vehicle per Reg 8.11.B only).
- 5.3.B.8** Provide details about the onboard charging system (Multi-Occupant vehicles only).

5.3.C Battery VDR

The battery VDR shall document the design of the solar car's battery system. All storage batteries used in the solar car must be approved by Headquarters. Energy capacity will be based on manufacturer's data. If an intermediate supplier is used, submit only the cell manufacturer's data as required on the Battery Approval Form. Note the definitions included in Reg 8.3.A. This report must:

- 5.3.C.1** Attach a Battery Approval Form from Appendix G for the main and supplemental pack.
- 5.3.C.2** Attach the battery specification data sheet obtained from each battery manufacturer.
- 5.3.C.3** Attach the Material Safety Data Sheet (MSDS) obtained from each battery manufacturer.
- 5.3.C.4** Attach a copy of the purchase receipt(s) confirming order quantity and that each battery model procured matches the data sheet provided.
- 5.3.C.5** Describe spill/damage protocols and procedures (if these are not provided in the MSDS then the team must obtain this information from the manufacturer and submit it to Headquarters with the MSDS).
- 5.3.C.6** Describe emergency procedures if the chase vehicle notices a fire in the solar car.
- 5.3.C.7** Describe the battery enclosure(s) and mounting provisions.

4.2 Be sure to remember to bring a minimum of three (3) individual battery cells (Reg 8.2) to Scrutineering for verification.

5.3.D Battery Protection VDR:

The battery protection VDR shall document the design approach used with respect to Reg 8.3. All batteries must be protected with technology appropriate to the chemistry used. This report must:

- 5.3.D.1** Provide school name and team number.
- 5.3.D.2** Attach Battery Approval Form from Appendix G for the main and supplemental pack.
- 5.3.D.3** Describe the battery pack configuration including cells per module, modules per string, strings in parallel, and total cell count.
- 5.3.D.4** Provide the over temperature setpoint(s) for each battery type (charge and discharge if different).
- 5.3.D.5** Provide the under voltage setpoint for each battery type.
- 5.3.D.6** Provide the over voltage setpoint for each battery type.
- 5.3.D.7** Provide the over current setpoint for each battery type (charge and discharge if different).
- 5.3.D.8** Provide a Block diagram for the BPS and components within the battery enclosures for each battery type.
- 5.3.D.9** Describe how the BPS will operate for start-up and fault conditions for each battery type.
- 5.3.D.10** Describe how firmware or settings will be rendered static and un-modifiable after inspection (i.e. sealed in battery enclosure).
- 5.3.D.11** Describe the functionality of driver dash and BPS fault strobe for External Cutoff switch, per Reg 8.7.C

5.3.E Solar Collector VDR

The purpose of this solar collector VDR is to pre-assess the vehicle's solar collector. All calculations must be reproducible by the Inspectors who will compare the report to the car that is presented at Scrutineering. This report will be based on the type of solar collector being used per Reg 8.1 and must:

5.3.E.1 Provide school name and team number.

5.3.E.2 Provide a complete list of cells used on your solar collector including the following information for each unique type of cell used:

- 1) Manufacturer's name and contact information.
- 2) Supplier's name and contact information.
- 3) US\$ cost per cell. Attach a copy of the purchase receipt(s) confirming order quantity and that each cell model procured matches the data sheet provided.
- 4) Cell type/chemistry.
- 5) Cell conversion efficiency.
- 6) Cell model or description.
- 7) Manufacturer's quote for cell area in cm².
- 8) Manufacturer's quote for cell power in Watts at STC.
- 9) Include a copy of the manufacturer's data sheets in the appendix of your report for each type of cell. *NOTE: Teams are only allowed a maximum of 3 types of solar cells. Any change in any of the cell characteristics would dictate a new type of cell being used.*

5.3.E.3 Provide a detailed breakdown of the measured dimensions of each type of cell used on the solar collector after trimming or cutting as placed on the solar car. Show the calculations for determining the cell area in cm².

5.3.E.4 Provide a detailed drawing showing the specific layout of each cell in the solar collector including how the solar collector is broken into sub modules and indicating which types and sizes of cells are used where on the solar collector.

5.3.E.5 Show the calculations for determining the total cell area for the whole solar collector in cm².

5.3.E.6 Show the calculations for determining the estimated power production of the whole solar collector in Watts at STC.

5.3.E.7 Indicate all other non-photovoltaic/photovoltaic technology used as defined in Reg 8. Report how these technologies will be used during driving and charging times. Describe how these technologies will fit into the car volume allowed per Reg 9.1.

Be sure to remember to bring a sample raw solar cell (Reg 8.1.F) of each type used to Scrutineering for verification.

5.4 Build Review

Each team must engage an experienced engineer or academic, responsible for reviewing and certifying that the solar car is designed and constructed using sound engineering practice, meets the design parameters where stated, and is roadworthy and fit for the purpose of being driven in the event(s). The form will be provided on the Team Status Board on the Event Website. The build review shall also include photographs of the vehicle in the state in which it was reviewed. Failure to submit this completed form, showing all items in full compliance, by the due date in Reg 4.2.A.7 shall result in a team dropping to the back of the queue for Scrutineering slots. Teams shall not begin the Scrutineering process before their completed Build Review is received.

5.5 Grading of Team Reports

Team documents will be reviewed by the Inspectors and a color grade will be given to each document on the Team Status Board with the following general meanings:

- **Green:** Requirement Completed. Submission is accepted; no further action required
- **Yellow:** Minor Action Required. Submission is insufficient; some revision to design or additional reporting is required. Resubmit with requested revisions
- **Red:** Significant Action Required. Submission is overdue, incomplete, or rejected. Significant revision to the design or significant additional reporting is required. Submit valid and full information ASAP

VDR submissions will not be reviewed before the team has a Green status on the relevant section of their PVDR. Green report status does NOT assure passing Scrutineering as not all aspects of a design can be fully reviewed during evaluation of written reports. All solar cars are subject to a detailed Scrutineering inspection at the Event.

6 Event Components

The dates and locations of each Event component shall be posted on the Event Website.

6.1 Scrutineering

6.1.A Acceptance at Scrutineering

Only teams who have obtained Green status on their Event Registration Submissions and who have paid the required Event fees will be accepted for Scrutineering.

6.1.B Participation at Scrutineering

Each team registered for the Event must submit their entry for inspection at Scrutineering prior to FSGP to verify compliance with these Regulations. In addition, spot checks for Regulation compliance may take place during and immediately after FSGP and ASC. The top five overall finishing cars in each class may be impounded immediately following the Event for a final inspection at the discretion of the Inspectors.

6.1.C Scrutineering Schedule

The inspection schedule will be announced prior to the start of Scrutineering with the inspection slot order randomly determined within the following framework. The first five teams to obtain all Green statuses on the Team Status Board will be given preferential inspection slots. At the time the Scrutineering Schedule is determined, VDR status for each station will be taken into account with Greens getting earlier slots, followed by Yellows, followed by Reds. Any teams who fail to present their Build Review on time will be given last priority inspection slots. Teams that fail to present their solar car at their designated inspection slots or fail to participate in mandatory team meetings may be given last priority for Scrutineering and risk not having enough time to complete the process.

6.1.D Scrutineering Format

Scrutineering will involve inspection stations including body & sizing, driver, electrical, battery protection, solar collector, mechanical, dynamic tests to verify handling and braking performance, and support vehicles. Instructions for Scrutineering and a detailed description of the Scrutineering tests will be distributed in advance to all registered teams.

6.1.E Configuration and Drivers for Scrutineering

All Drivers must be present for designated Scrutineering inspection stations. The driver selection and car configuration are at the discretion of the Inspectors for each station. Teams may be required to repeat tests with different drivers and/or configurations as directed by the Inspectors. Different numbers of occupants qualify as different configurations for Multi-Occupant Vehicles.

6.1.F Scrutineering Status

Teams will be graded after each Scrutineering station inspection with a status color as follows:

- (1) Green – Team has passed station
- (2) Blue – Improvement needed / cleared for track with penalty condition
- (3) Yellow – Improvement needed / cleared for dynamics testing
- (4) Red – Team has failed station / safety issue must be addressed
- (5) Grey – Not Applicable to Team

6.2 FSGP

FSGP is a closed-course track competition. Reg 14 outlines the format for FSGP.

6.2.A Acceptance at FSGP

Only teams who have obtained Green or Blue status for each Scrutineering Inspection Station will be accepted into the FSGP competition.

6.3 ASC

ASC is a cross-country Tour. Reg 12 outlines the format ASC.

6.3.A Acceptance at ASC

Only teams who have paid the full Event fee, obtained Green status for each Scrutineering Inspection Station, and successfully participated in FSGP meeting the Qualification Requirements from Reg 14.12 will be accepted into the ASC competition. Blue Scrutineering status can be changed to Green by resolving the underlying issue or accepting the penalty condition prior to the start of the Tour.

7 Vehicle Classes

Three (3) classes of solar vehicle will be recognized as part of the Event:

- Single-Occupant
- Multi-Occupant
- Demonstration

7.1 Vehicle Class Definitions

7.1.A Single-Occupant (SOV)

This class comprises solar powered vehicles designed for a single-occupant. These vehicles are akin to the “Challenger” class of vehicles from the World Solar Challenge.

7.1.B Multi-Occupant (MOV)

This class comprises solar powered and grid-charge vehicles designed for multiple-occupants. These vehicles are akin to the “Cruiser” class of vehicles from the World Solar Challenge.

7.1.C Demonstration (DEMO)

7.1.C.1 Vehicles in this class must have entered, passed Scrutineering, and qualified for ASC, FSGP or another recognized solar car event within four calendar years prior to the current Event, OR, they must be designed to released Regulations for an upcoming solar car event.

7.1.C.2 At Scrutineering, vehicles in this class will be inspected to the Regulation set they were designed to and will be subject to additional regulations associated with safety from the Single-Occupant and Multi-Occupant class vehicle technical Regulations.

7.1.C.3 This class is Demonstration only. Vehicles in this class will not be scored or ranked.

SECTION 2 – TECHNICAL

8 Power

Natural solar radiation received directly by the solar collector is the only source of energy that is allowed for propulsion, except for energy stored in the solar car's battery system at the beginning of the Event or metered AC charging of MOVs. Energy recovered from the motion of the car during Track or Tour hours may also be used.

8.1 Solar Collector

8.1.A Cell Type

Only commercially available silicon based photovoltaic cells are allowed. Silicon-perovskite tandem cells are considered to be silicon based.

8.1.B Solar Collector Size Limits

The solar collector cannot exceed a maximum of 6 m² for Single-Occupant class solar cars, and 5 m² for Multi-Occupant class solar cars.

8.1.C Definition of Area

Solar collector area is calculated by summing the total area of each cell (including all exposed bus bars, junctions and internal structure) from manufacturer data sheets, and verified by measurements.⁴

8.1.D Concentrators

If the solar collector consists of photovoltaic cells used with concentrators such as reflectors or lenses then the total aperture of the solar collector must not exceed the allowable total area for non-concentrator photovoltaic solar collectors. Teams wishing to use concentrator photovoltaic solar collectors must contact Headquarters for more information.

8.1.E Maximum Number of Cell Types and Sizes

Teams are limited to no more than three (3) types or sizes of solar cells.

8.1.F Validation Documentation

At Scrutineering, teams must provide sample cells of each type and size installed on the vehicle as well as a detailed map of the vehicle solar collector for validation per Reg 5.3.E.

8.1.G Solar Collector Connection and Stands

8.1.G.1 All portions of the solar collector, physical attachment to the solar car, and all electrical connections between the solar collector and the solar car must be carried by the solar car. This includes but is not limited to stands, supports, and cables. This does not include hand tools that do not become part of the solar collector structure.

8.1.G.2 Stands used for pointing the solar collector while the car is stationary must be self-supporting; stands that rely on people as a means of support will not be allowed. Teams will be required to demonstrate stands at inspection.

Team members may temporarily steady stands in inclement weather, but it will be obvious to Officials if human support of the solar collector becomes routine.

8.1.H Water Spray

Spraying water (or other fluids) on the solar collector between battery un-impound in the morning and battery impound in the evening is not allowed.

8.1.I Solar Collector Reconfiguration

The solar collector can only be reconfigured for charging when the solar car is stationary.

8.1.J External Irradiance Amplification

External devices intended to increase the irradiance on the solar collector or increase the efficiency of

⁴ For the purposes of these Regulations, the area of an uncut Maxison cell having a width and length of 125 mm and diameter of 160 mm is 153.33 cm². An uncut Maxison cell having a width and length of 125mm and a diameter of 166 mm is 155.06 cm².

the solar collector must not be used at any time. Ground sheets used when charging must not increase the irradiance on the solar collector.

8.2 Energy Storage

Solar cars are allowed to store solar-generated energy in an energy storage system composed of commercially available individual cells. The storage system is limited by the rated energy capacity on the battery manufacturer's datasheet. If the datasheet provides charge capacity (Ah) instead of energy capacity (Wh), nominal battery voltage will be used to convert to energy capacity.

Adherence to energy capacity limits does not imply automatic battery approval. The Battery Approval Form from Appendix G must be submitted to Headquarters before official approval may be issued. The Inspectors reserve the right to refuse approval of modules. A minimum of three (3) unaltered samples of individual cells must be furnished for verification during Scrutineering for the main battery.

8.2.A Battery Energy Capacity Limits

8.2.A.1 Single-Occupant solar cars are limited to 5.25 kWh of main storage capacity.

8.2.A.2 Multi-Occupant class solar cars are limited to 15.5 kWh of main storage capacity.

8.2.B Other Energy Storage Methods

Other energy storage technologies not mentioned (such as other battery technologies, ultracapacitors, flywheels, or fuel cells) will need to be evaluated by Headquarters. Samples and details of proposed systems must be submitted before the date in Reg 4.2.A.2.

8.2.C Supplemental Batteries

8.2.C.1 Supplemental, replaceable batteries carried in the solar car are allowed to power: Main Power Contactors, driver ventilation fans (if solely used for driver ventilation), telemetry, and the horn. All other systems present on the car must be powered off the main battery.

8.2.C.2 Supplemental battery power must be used to momentarily power the battery protection system (BPS) as defined by Reg 8.3 to verify safe battery parameters during start up before energizing the Main Power Contactors.

8.2.C.3 During a BPS fault in which the Main Power Contactors have automatically opened, the supplemental battery must be used to power the BPS, BPS Strobe, and BPS Fault Dash Indicator. While in such a fault state, the supplemental battery may power any other systems (signal lights, rear view camera, telemetry, battery ventilation, etc) that do not provide motive power to the vehicle.

8.2.D Batteries Internal to Commercial Devices

Commercially available panel meters, cell phones, tablets, driver communication radios, and other electronic devices with their own batteries are allowed, provided that power is not exchanged with the vehicle electrical system.

8.3 Protection Circuitry

All batteries must have protection circuitry appropriate for the battery technology used. Proof is required at Scrutineering that the protection system is functional and meets manufacturer's specifications. The test procedure can be downloaded from the Event Website, and the protection system design shall allow for such testing. All protection circuitry shall be contained in the battery enclosures per Reg 8.4.

8.3.A Definitions

- 8.3.A.1 Cell:** The smallest available source of energy in the battery pack as purchased from a manufacturer. A single electrochemical cell.
- 8.3.A.2 Module:** The smallest grouping of paralleled cells easily removable in a battery pack.
- 8.3.A.3 String:** The smallest series of modules that make-up the battery pack required voltage.
- 8.3.A.4 Protection Limit:** The measured level determined to be adequate to protect from an event.
- 8.3.A.5 Active Protection:** System in which measurements are constantly monitored and where actions are taken immediately without operator intervention to open the Main Power Contactors should a BPS fault occurs. Any BPS faults must latch such that a manual clearing process is required by the driver after faults have been verified clear by the protection system.⁵
- 8.3.A.6 Passive Protection:** System in which measurements are monitored by the driver and where action is driver controlled.
- 8.3.A.7 Battery Protection System (BPS):** The system that applies Active Protection to protect the battery pack.
- 8.3.A.8 Tap:** Wiring that is bonded to an electrical connection point in a circuit for remote voltage or signal measurement
- 8.3.A.9 Safe State:** A stable shut down condition where all internal electrical loads connected to the high voltage battery inside the battery enclosure are either isolated or in a high impedance state/off and all external electrical loads are completely disconnected by isolating the high voltage battery pack from all high voltage conductors exiting the battery enclosure.

8.3.B Types

- 8.3.B.1 Main:** All main battery packs must have Active Protection such that over-voltage, over-temperature (for charge and discharge rating), over-current and under-voltage trigger a BPS fault per Reg 8.3.A.5. The level of protection measurement is required down to the module level at a minimum and may be required at a cell level depending on the cell manufacturer. The fuses required in Reg 8.6 are not sufficient for battery over-current protection. Chemistries other than lithium may be allowed with less protection at the discretion of the inspectors
- 8.3.B.2 Supplemental:** Supplemental batteries may be either 1. constructed of only primary cells, 2. lead-acid with passive protection and not charged in the vehicle, 3. lead-acid with active protection, or 4. commercially available and packaged from the manufacturer with an integrated, active battery protection system.

⁵ The Battery Protection System is intended to be the last level of protection for the battery. Teams should implement soft limits where applicable to prevent BPS faults to minimize or eliminate the need to stop to investigate/restart during the event.

8.4 Battery Enclosures

All registered and sealed battery modules, supplemental batteries, battery protection circuitry per Reg 8.3, and main fuses per Reg 8.6 must be fully contained in enclosures that are electrically isolated from the solar car. The inside of the enclosures must be a non-conductive material. No more than two separate such enclosures are allowed for the main and supplemental batteries.

8.4.A Isolation

The resistance measured between the battery terminals and any portion of the solar car chassis shall be greater than 100 ohms per 1 volt of maximal battery potential.

8.4.B Mounting

The battery enclosures must be secured to the solar car chassis so as to prevent them or the modules within from coming loose in the event of an accident or rollover. Consider a 5G impact in the forward or vertical direction. Nylon luggage type buckles are not acceptable means of securing the enclosure.

8.4.C Marking

The top of each battery enclosure must be marked using letters at least 10mm high with "Caution: Chemical Hazard" and "High Voltage" and any other standard hazard markings specific to the type of battery enclosed. The type (i.e. Li-ion) of the battery must be marked on the top of the battery enclosures(s) in 10 mm high letters.

8.4.D Ventilation

The battery enclosure(s) containing the main battery must be equipped with a forced ventilation system. It must operate whenever the battery system is enabled and the Main Power Contactors are closed. The ventilation system must pull air through the battery rather than push it – ie, the battery enclosure should be at negative (not positive) pressure relative to ambient pressure. The exhaust airflow must be directed through sealed ducting to the exterior of the solar car in a location away from any airstream that may reach the occupant(s).

8.4.E Security

To preclude unauthorized access to the battery/enclosure, a seal will be placed to indicate contravention of this regulation. Provisions shall be made to seal the battery/enclosure by the team. Should access to a "sealed" battery/enclosure be needed, the team needs to inform an Official of their intent to access the battery/enclosure, and request the Official to log the activity and retain the seal. Battery exchanges will not occur during the Event without Inspector support.

8.5 Cooling

All energy used to cool any of the vehicle's electrical equipment (battery, solar collector, motor controller, etc) must be derived from the solar collector. All cooling systems must be contained within the solar car. "Open cycle" cooling methods that rely on energy sources or sinks not derived from the car's solar collector are not allowed, with the exception of cooling of the battery in an emergency and with the main contactors open. See also Regs 8.1.H and 8.4.E.

For instance, you can not pack ice around your motor controller unless the ice was made in an ice maker carried within and powered by the solar car.

8.6 Fusing

8.6.A Main

A DC-rated fuse must be placed in series with the battery within each battery enclosure. Both leads to the fuse must be mechanically constrained. The fuse rating must not exceed 200% of the maximum expected current draw or 75% of the rated wire current capacity. The fuse must be rated in excess of the maximum DC voltage of the main battery pack, must be rated to break the fault current due to a shorted pack, and must be rated to protect the Main Power Contactor.

8.6.B Branch

All other wiring branching off the main bus circuit must have properly sized fuses. Fuses must be located near the branch point, either within the same enclosure or before a reduction in rated conductor ampacity.

8.6.C Cell Voltage Taps

8.6.C.1 When in the Safe State, residual current draw on any cell measurement tap shall be less than 10 mA.⁶

8.6.C.2 Cell voltage taps may not leave the battery enclosure(s).

8.7 Power Switching

8.7.A Main Power Contactors

8.7.A.1 Solar cars must be equipped with normally open and non-latching Main Power Contactors that are located within the main battery enclosure(s).

8.7.A.2 The solar car driver must have overriding control and the ability to switch the Main Power Contactors off at all times.

8.7.A.3 The location of the switch that controls the Main Power Contactors must be within easy reach of the driver and clearly labeled.

8.7.A.4 In the off position, the Main Power Contactors must electrically isolate the high voltage battery, motor, and solar collector from each other and put the solar car in the 'Safe State'. The high voltage battery pack contactors must be electrically located first in series after the battery terminals, or second in series after the main fuse.

See additional regs interpretation in Appendix J

8.7.A.5 The Main Power Contactors must be DC rated and capable of interrupting an overcurrent condition.

8.7.A.6 The BPS must have overriding control and the ability to switch the Main Power Contactors off at all times in the event of a BPS fault.

8.7.B BPS Fault Dash Indicator

The driver is to have an illuminated dash indicator for a BPS fault to provide warning of an automatic opening of the Main Power Contactors. This indicator must be labeled in text at least 5mm high and both the indicator and label must be within the driver's direct line of sight when the driver is seated in their normal driving position.

⁶ Storage of batteries should include regular charging and removal of all parasitic loads to prevent battery damage

8.7.C External Power Cut Off Switch

For emergency use, an external power cutoff switch must be present on the exterior of the solar car with overriding control and the ability to switch the Main Power Contactors off at all times.

8.7.C.1 Location: The switch location shall be on the solar car's exterior near the cockpit, on the same side from which the driver's egress is opened. The switch must be on an upward-facing surface; it must not be on surfaces that are angled such that the surface normal is pointing below the horizon.

8.7.C.2 Operation: Only "Push" switches are allowed. The actuator must be colored red, and at least 20mm in diameter (non-round push buttons must be large enough to fit an inscribed 20mm circle). Actuation of the external switch must illuminate the BPS Fault Indicator on the exterior of the car and BPS Fault Dash Indicator for the driver. The switch shall be a latching type switch.

8.7.C.3 Marking: The external actuator must be clearly marked by the international marking of a red spark within a white-edged blue equilateral triangle with a minimum side length of 150 mm. In addition, clear directions on how to operate the actuator must be displayed using letters 10 mm tall or larger.

8.7.C.4 Covering: The external actuator may be covered with a colorless transparent cover. Without the use of tools or significant force, it must be demonstrated that the actuator may be easily operated through the cover. The cover must be labeled in such a manner as to how to operate the actuator through the cover. The blue triangle marking may be located on the cover, but must not obstruct the view of the actuator.

8.8 Cable Sizing

All electrical cables must be properly sized for expected system currents.

8.9 Control

8.9.A Sole Control

Any parameter that influences the safe and reliable operation of the vehicle must be under the sole control of the driver.

8.9.B Accelerator

Accelerator mechanisms on solar cars must be free moving, and when released, must return to the zero position. Should a pedal accelerator be used, it shall be mounted such that it is operated by the right foot and it shall be located to the driver's right of the brake pedal (if equipped).

8.9.C Cruise Control

If the solar car is equipped with cruise control, it can only be activated by the driver. The cruise control must be designed to automatically deactivate when the brake controls are manually actuated by the driver, or when the car is shut off.

8.9.D Reverse

The solar car must be able to be driven backwards under its own power with the driver seated in the normal position.

8.10 Electrical Shock Hazards

All exposed or easily exposed conductors, junction boxes, solar cells, etc., operating at greater than 60V must be protected from inadvertent human contact and must be marked "High Voltage" in letters at least 10 mm high.

Exposed carbon fiber is conductive so care should be taken to ensure electrical components are isolated from it. Any covers allowing access into the enclosures must be firmly secured.

8.11 Impound

Battery enclosures shall be designed such that they can be Impounded per Regs 12.17 and 14.13.B. Enclosures may either be removed from the vehicle and placed in an impound box or impounded in the vehicle.

8.11.A Impound Box

Teams utilizing the impound box option shall provide an impound box that fully contains the battery enclosure(s) per Reg 8.4 in such a way that completely restricts access to the battery enclosure(s).

8.11.A.1 The impound box must allow for an Official to secure it closed with a maximum of two (2) locks/seals.

8.11.A.2 The impound box must be constructed such that it does not contain external hardware or hinge pins that can be removed to access the battery enclosure without breaking the seal(s).

8.11.A.3 The top of the impound box must clearly be marked using at least 10 mm high letters with team number and university name.

8.11.B Impound In Vehicle

Teams utilizing the impound in vehicle option shall provide a robust impound solution that allows the event organizers to lock/seal any high voltage battery power connectors/conductors.

8.11.B.1 The solution shall seal any main battery high voltage power connectors/conductors such that the team cannot charge the battery from external energy sources. This includes the battery enclosure lid(s), and other openings such as air inlets/outlets that could allow access to high voltage connections, as well as any power connections to the motor, solar collector, etc.

8.11.B.2 The solution shall have provisions such that an Official can secure the main battery system with a maximum of four (4) locks/seals used in the daily un-impound/impound process. Additional locks/seals may be applied during Scrutineering so long as they do not need to be broken every time the battery is impounded or released from impound.

8.11.B.3 The battery enclosure(s) shall be constructed without any external hardware or hinge pins that could be removed to access the battery enclosure(s) without breaking the seal(s).

8.11.C Impound Security Seal Marking

Each impound security seal location must be marked using 10 mm high letters with "Seal [#] of [total]". For example: "Seal 2 of 3".

8.12 Multi-Occupant Vehicle Charging and Metering

8.12.A Charger

MOVs must be equipped with a J1772 or NACS capable onboard vehicle charger that accepts external energy from a 120-240Vac power source and outputs DC power with an adjustable current limit to charge the main battery pack. The AC input side of the charger must include a safety ground connection from the vehicle power inlet and this must be tied to the charger/chassis if it contains any exposed metal. The charger must be rigidly secured in the vehicle in a location not susceptible to the ingress of water. The DC output of the charger must be electrically isolated from the AC input. The inputs and outputs on the charger must also be electrically isolated from the vehicle chassis.

8.12.B Vehicle Power Inlet

MOVs must be equipped with a J1772 or NACS vehicle power inlet receptacle mounted and secured to the vehicle for delivering power to the onboard charger. The vehicle charging system must function in a manner that is compliant with the SAE J1772 or NACS automotive standard. This standard requires a communication handshake between the Electric Vehicle Supply Equipment (EVSE) and the onboard vehicle charger before charging can occur.

8.12.C Charging Adapter

For MOVs designed for use in geographies with other standard EV connectors, an appropriate adapter may be used to go between a J1772 or NACS plug and a different vehicle power inlet receptacle. In this case, the charging adapter must be no longer than 1 m in length and must be carried in the MOV while not in use.

8.12.D Energy Metering

Multi-Occupant vehicles will be required to carry an onboard AC energy meter provided by the Event Organizers – see details in Appendix I. The meter will have a wide measurement range for charging at 120-240Vac and up to 40A. The meter enclosure is a 6.7" x 10.63" x 4.33" box which must be installed inline between the J1772 or NACS vehicle power inlet and the AC input on the onboard vehicle charger. The energy meter box will feature a NEMA 14-50P plug input and a NEMA 14-50R output receptacle. The location of the meter in the vehicle must be such that its display can be read and teams must keep photo records with timestamps of the kWh meter reading at the beginning and end of each external energy charge.

The main battery enclosure must provide a dedicated power port and a dedicated charger contactor that is utilized only for external energy charging. All exposed connectors/conductors on the AC/DC charging power lines between this battery enclosure port and the J1772 or NACS vehicle power inlet must be enclosed in an approved manner that can be locked/sealed by the Event Organizers to prevent any of these connections from being unplugged or tapped into to bypass the energy meter or the J1772 or NACS power inlet. The team must also provide a way to seal their charger such that unauthorized access to its internal components is prevented.

8.12.E Charging Safety

Any covers for AC or DC power connections must be non-conductive and only removable with the use of tools. Such covers must be marked using 10 mm high letters with "Caution: High Voltage". Transparent covers that allow for visibility of connections are preferred. All conductors used for MOV charging must be appropriately sized to carry the vehicle's maximum possible charge current.

Charging from an external energy source is only allowed when the vehicle is stationary. When a J1772 or NACS plug is connected, the vehicle must automatically prevent the use of the drive motor(s).

When charging from an external energy source the BPS must always actively monitor the battery state. In case of a BPS fault condition where the Main Power Contactors controlled by the BPS open, the output of the MOV charger must also disconnect from the battery pack. To minimize BPS faults, the onboard charger should curtail charging automatically when the main battery pack approaches a full charge.

8.12.F Electric Vehicle Supply Equipment (EVSE)

Teams are responsible for providing their own Level 1/Level 2 SAE J1772 or NACS Electric Vehicle Supply Equipment to connect from an external energy source (utility grid, generator, etc.) to the vehicle power inlet. It is recommended for teams to bring adaptors that will allow them to charge from various North American power receptacles. The most common receptacle is the 120 Vac NEMA 5-15R but it is also recommended to have 240Vac adaptors available for faster charging from receptacles such as the NEMA 14-50R, NEMA 14-30R, NEMA 6-30R. Teams are not required to carry their J1772 or NACS Electric Vehicle Supply Equipment in the solar car.

Note: The Event Organizers are not responsible for providing external energy sources for MOV teams to use during the event or any adapters required to charge your vehicle. It is recommended that MOV teams have a generator available that could be used for charging the vehicle in the event that an acceptable alternate power source can't be identified.

8.12.G MOV Impound

In impound configuration, the metered J1772 or NACS vehicle power inlet must be the only functional power connection to the main battery pack. Failure to properly impound during the designated hours at the Event will result in a full unmetered external energy charge being added in the team's Score.

9 Solar Car Body

9.1 Solar Car Dimensions

The solar car (including solar collector) must not exceed the following maximum dimensions when moving under its own power:

- Length = 5.80 m
- Height = 1.65 m
- Width = 2.30 m

When turning corners, wheels and wheel fairings may exceed these dimensions.

9.2 Driving Configuration

While the vehicle is moving under its own power, reorientation and reconfiguration of wheel fairings and other aerodynamic devices is allowed, however, reorientation or tilting of the solar car body is prohibited.

9.3 Ground Clearance

Solar cars may have adjustable ride height systems. The car must have a configuration where the fully-laden solar car (except wheels, tires, and wheel hubs) has a ground clearance greater than 75 mm, and must be able to raise itself into this state from lower height states without the use of tools, team members other than the driver, etc.

9.4 Lighting

Solar cars must have lighting as outlined below.

All lights required in this regulation shall be at least as bright as the reference standard defined in Appendix C from 30 m away throughout the required viewing angles as defined below. Inspection for the viewing angles and light intensity can be conducted at any point within the defined distance and view angles. All positions are to the outermost light emitting portion of the light.

Inspect your lights outdoors, in bright sunlight, at the prescribed distances and angles, PRIOR to arriving at the event! Lights that look great inside dim garage bays may be essentially invisible in real-world outdoor conditions.

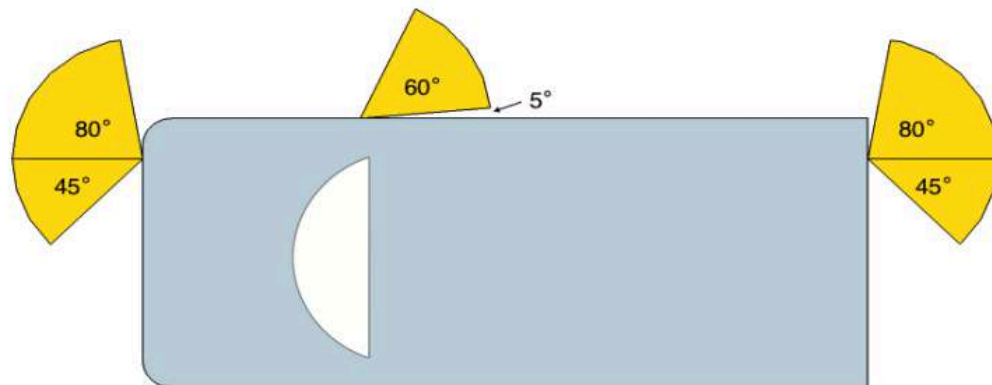


Figure 1: Turn Indicator Horizontal Visibility (for the right turn indicators)

9.4.A Front Turn Signals

- 9.4.A.1 Position:** shall be located at least 600 mm apart (at least 400 mm apart if the car is less than 1300 mm wide) and at least 350 mm above the ground.
- 9.4.A.2** Shall flash between 60 and 120 pulses per minute
- 9.4.A.3 Color:** amber
- 9.4.A.4 Viewing Angle:** 80° from center in outwards, 45° from center in inwards (see Figure 1), 5° down, and 15° up from horizontal.

9.4.B Rear Brake Lights

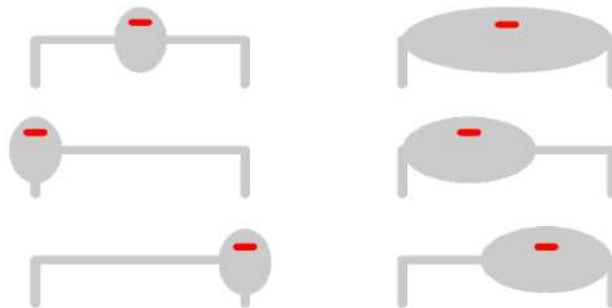
- 9.4.B.1 Position:** shall be located within 400 mm of the extreme outer edge of the car on each side, at least 600 mm apart (400 mm if the car is less than 1300 mm wide), and at least 350 mm above the ground.
- 9.4.B.2** It is permissible to have one set of lights per side of the car which operate as both the brake lights and turn indicators. The turn indicator operation has the priority in operation.
- 9.4.B.3 Color:** red
- 9.4.B.4 Viewing Angle:** 45° from center outwards and inwards, 5° down, and 15° up from horizontal.
- 9.4.B.5 Operation:** Brake lights shall operate if and only if the driver presses the brake pedal or regenerative braking is active.

9.4.C Rear Turn Signals

- 9.4.C.1 Position:** shall be located within 400 mm of the extreme outer edge of the car on each side, at least 600 mm apart (400 mm if the car is less than 1300 mm wide), and at least 350 mm above the ground.
- 9.4.C.2** It is permissible to have one set of lights per side of the car which operate as both the brake lights and turn indicators. The turn indicator operation has the priority in operation.
- 9.4.C.3** Shall flash between 60 and 120 pulses per minute
- 9.4.C.4 Color:** amber (if separately equipped from Rear Brake Lights)
- 9.4.C.5 Viewing Angle:** 80° from center outwards, 45° from center inwards, 5° down, and 15° up from horizontal.

9.4.D High Mounted Center Brake Light

- 9.4.D.1 Position:** Viewed from behind the solar car, the lateral position of the light shall coincide with the visual center of the solar car (see the examples in the following diagram). The top of the lamp shall be less than 150mm below the highest fixed point of the car, and the bottom of the lamp shall be higher than the top of the rear brake lights.



- 9.4.D.2 Color:** red
- 9.4.D.3 Viewing Angle:** 10° from center in both left and right, and 10° up and 5° down from horizontal.

9.4.E BPS Fault Indicator Light

- 9.4.E.1 Position:** Located adjacent to the center brake light, see Reg 9.4.D.1. May be on any side (left, right, top, or bottom) of the center brake light.
- 9.4.E.2 Activation:** The fault light shall activate within 3 seconds of any fault where the Battery Protection System (BPS) actuates to automatically open the Main Power Contactors and remain active while the BPS fault is present. The fault light shall also activate within 3 seconds of the actuation of the External Power Cutoff Switch (Reg 8.7.C.2). The fault light shall not activate under any other conditions.
- 9.4.E.3** Shall flash between 60 and 120 pulses per minute
- 9.4.E.4 Color:** white strobe
- 9.4.E.5 Viewing Angle:** 10° from center in both directions and 10° up from horizontal.

9.4.F Emergency Hazard Lights

The front turn indicators, side marker turn indicators, and rear turn indicators shall be able to be activated simultaneously and flash in sync as an Emergency Hazard signal.

9.4.G Side Marker Turn Signals

- 9.4.G.1 **Position:** shall be mounted on each side of the vehicle between 500 and 1800 mm rearward from the absolute front of the vehicle and within 400 mm of the extreme outer edge of the car.
- 9.4.G.2 Shall flash between 60 and 120 pulses per minute
- 9.4.G.3 **Color:** amber
- 9.4.G.4 **Viewing Angle:** 5° down / 15° up from horizontal, horizontal visibility as shown in Figure 1

9.5 Horn

Solar cars shall be equipped with a horn that can be heard at a sound power level between 75 and 102 dBA at a distance of 15 m forward from the absolute front of the vehicle. The horn shall be permanently mounted, operated from the steering wheel, and shall be able to operate for up to 5 minutes continuously at the required volume.

9.6 Outside Air Circulation

Outside air, from intake vents and directed towards the occupant's face, must be provided. Should intake vents from the wheel openings be used, the natural air flow rate through the ducting to the occupant compartment shall be augmented by a ventilation fan.

9.7 Visibility

9.7.A Eye Height

In the normal driving position with a fully laden solar car, all occupant's eyes must be at least 700 mm above the ground.

9.7.B Windshield

- 9.7.B.1 All solar cars must have a windshield that is securely mounted to the solar car.
- 9.7.B.2 The windshield must be made of shatter-resistant material; the material must have a Notched Izod Impact Strength of at least 30 kJ/m² (ISO 180/1A) or 320 J/m (ASTM D256).
- 9.7.B.3 The windshield must be free of excessive distortion.
- 9.7.B.4 The windshield shall not be tinted to the extent that the driver cannot be clearly observed from outside the solar car.
- 9.7.B.5 The solar car driver must be able to discern traffic light colors through the windshield.

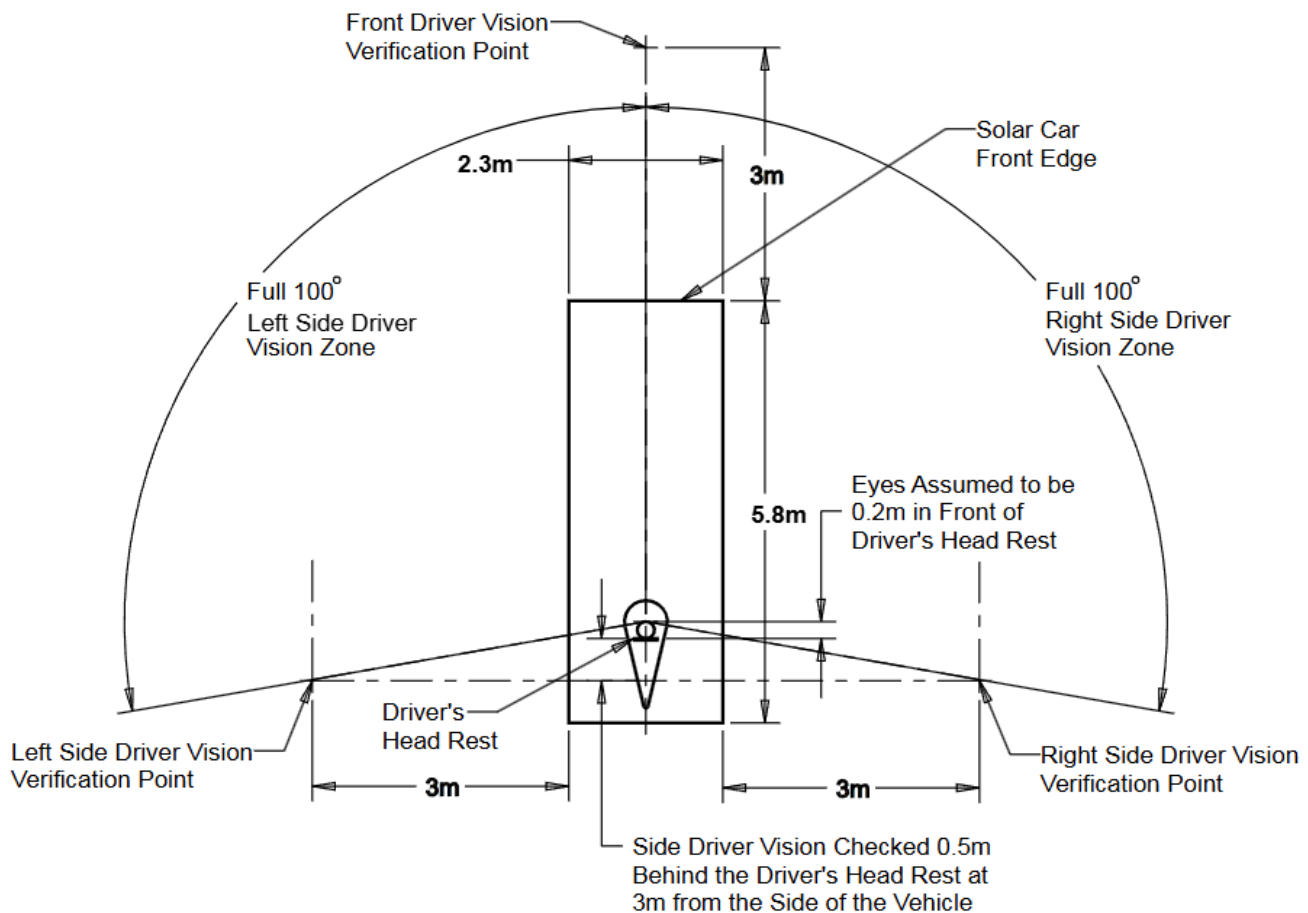
9.7.C Rain Clearing

- 9.7.C.1 Solar cars must have a method to clear the windshield from any falling rain such that the vision requirements of Reg 9.6.B can be met.
- 9.7.C.2 The clearing method must be operable at all times and must be in use when it becomes necessary to use the windshield wipers on the team's support vehicles.
- 9.7.C.3 Hydrophobic coatings (such as Rain-X) are acceptable.

9.7.D Forward and Sideward Vision

From the normal driving position, the driver must be able to see at all times, without artificial assistance, points at the following locations:

- 9.7.D.1** A point on the ground 8m ahead of the absolute front of the vehicle, aligned on the centerline of the solar car.
- 9.7.D.2** A point 6.4 m above the ground and 12.2 m ahead of the absolute front of the vehicle, aligned on the centerline of the solar car. This may be inspected with an object lower and closer to the car in order to simulate this viewing angle.
- 9.7.D.3** The driver will be required to identify 50mm high letters held about 1 m off the ground at the three verification points indicated in the image below. Spot checks may be made with these letters at distances 3m from the edge of the car throughout the entire left and right driver vision zones indicated. Some elements of the roll cage may obstruct a portion of these zones, however, the driver's view must be unobstructed as much as is reasonably possible by the solar car structure.



9.7.E Rear Vision

- 9.7.E.1** All solar cars must be equipped with a rear-view system that at all times will allow the driver to see a vehicle 15 m directly behind the solar car and up to 30° off center. The system must provide the driver with a single reflex type image and must operate without driver input. The driver will be required to identify the direction of an arrow with a 200 mm thick brush stroke on a 1 m² board held about 1 m off the ground.
- 9.7.E.2** If equipped: the camera and view screen shall be fixed in position such that road bumps and vehicle vibration will not alter the viewing angles.
- 9.7.E.3** If equipped: the view screen shall be positioned such that the driver shall be able to see the view screen while seated in normal driving position.

9.8 Egress

9.8.A Performance Requirement

- 9.8.A.1 All teams shall define a primary egress direction for their solar car.
- 9.8.A.2 Multi-Occupant teams shall also define a secondary egress direction. The primary and secondary directions must be separated by at least 90 degrees, and both primary and secondary directions cannot be on the same side of the solar car. The primary and secondary directions may be different for each occupant.
- 9.8.A.3 For Single-Occupant solar cars, teams will be required to demonstrate that the occupant can exit the vehicle unassisted, standing clear of the plane of the car, in less than 10 seconds
- 9.8.A.4 For Multi-Occupant solar cars teams will be required to demonstrate that each occupant can exit the vehicle unassisted, standing clear of the plane of the car, in less than 10 seconds for the occupant's primary direction and less than 15 seconds for the occupant's secondary direction⁷. It is envisioned that during the testing the drivers and occupants will be tested at the same time.
- 9.8.A.5 The solar car shall not be chocked during the egress test and the test shall start with the parking brake disengaged.

9.8.B Occupant Cockpit and Opening

- 9.8.B.1 The occupant's cockpit must be designed to allow each occupant to exit the vehicle unassisted.
- 9.8.B.2 Occupant egress openings must be able to be secured and released from both the inside and outside of the vehicle. Such openings must not be sealed or secured with adhesive tape at any time.
- 9.8.B.3 Occupant egress openings must be **positively latched closed** – Velcro, magnets, cabinet catches, and similar closure methods that do not positively retain the egress opening will not be allowed.
- 9.8.B.4 Egress openings, and the methods of opening, must be clearly indicated on the exterior of the solar car, and be visible and achievable by a single first responder not familiar with the solar car.

Instructions may include diagrams

- 9.8.B.5 The external egress opening release must be located within 300mm of the edge of the egress opening.

⁷ It is envisioned that during Scrutineering the egress of the drivers and occupants will be tested at the same time. I.e. the driver (assumed to be at the front left position of the car) will egress in their primary side, (i.e. assumed to be the left front door), whereas the passenger (assumed to be at the front right position of the car) will egress to their secondary side (i.e. assumed to also be the left front door). This is intended to check egress for a scenario where the car is against an object on the right preventing egress in that direction.

9.9 Ballast

The official mass of each occupant, including clothes (including shoes, excluding helmet, with empty pockets), will be 80 kg. If an occupant is lighter than 80 kg, ballast will be added to make up the difference. If an occupant is heavier than 80 kg, no credit will be given.

9.9.A Ballast Bag

Each registered solar car occupant will be allowed one bag to contain their required ballast. This bag will be a single colored canvas bank (coin) bag provided by the Event Organizers with dimensions of 305 mm x 482.5 mm. Ballast must be able to be contained within the canvas bag allowing security seals to be applied. Consideration should be made to ensure that a full ballast bag will fit securely in the car's ballast box(es).

9.9.B Ballast Box

9.9.B.1 Each solar car must have one (1) ballast box for each occupant.

9.9.B.2 Each occupant's Ballast Box shall be located within a 300 mm horizontal distance of the occupant's hip location.

9.9.B.3 Each Ballast Box shall have a lid which is secured closed for carrying ballast. The Ballast Box(s) must be securely fastened to a structural member of the solar car and/or be demonstrated to hold the ballast fixed in the event of an impact.

As ballast may be several 10s of kilograms of mass, and you don't want this mass flying around inside the car unsecured in a crash, don't let your ballast box be an afterthought. Ballast boxes zip-tied to the chassis will not pass inspection.

9.9.C Common Ballast

9.9.C.1 Teams entered in the Single-Occupant class may elect to use a Common Ballast. The Common Ballast provision does not apply to drivers or passengers in MOVs.

9.9.C.2 Should a team elect to use a Common Ballast, then each solar car driver shall have one (1) individual ballast bag and the Common Ballast bag. The sum of the two (2) ballast bags shall be equal or greater than the ballast required to bring the driver's mass up to the mass specified in Reg 9.9.

9.9.C.3 Teams that plan to use a Common Ballast must equip their cars with a Common Ballast Box that complies with Reg 9.9.B.3. This box may be located anywhere within the vehicle. The Common Ballast bag must be able to be secured inside the Common Ballast Box with a security seal at the start of the event.

9.9.D Ballast Access

Occupants and their corresponding ballast will be identified with unique identification tags. The tags on the ballast carried by the solar car must match the tags on the occupant at all times. The ballast bag and its identification and security markings must be visually accessible by the Event Officials during driver changes.

9.9.E Ballast Type

Teams will provide their own material for ballasting purposes. Ballast types allowed shall be either steel shot, lead shot, or coin. Other ballast types may be allowed at the discretion of the Event Officials. Consideration should be made with respect to the density of material selected and a driver's mass to ensure that the required ballast needed will fit into the bag provided.

9.10 Solar Car Markings

9.10.A Team Numbers

- 9.10.A.1** Each team registered for the Event will have a unique number approved by Headquarters (positive integer, three (3) digits maximum).
- 9.10.A.2** This number must be clearly displayed on both sides of the solar car and clearly visible from a distance of 3 m perpendicular to the side of the vehicle at a viewing height of 1.8 m above ground.
- 9.10.A.3** Each number must have a minimum of 50 mm of unobstructed background color on all sides. These colors can be black on white, white on black, or another high-contrast color approved by Headquarters. The numerals themselves must be a minimum of 250 mm high, 120 mm wide (except the numeral one), and have a minimum solid color brush stroke of 40 mm. Numbers containing more than one digit must have a minimum of 25 mm spacing between the digits.

9.10.B Institution Name(s) & Sponsors

- 9.10.B.1** The name of the team's host Institution(s) must be clearly displayed on both sides of the solar car and clearly visible from a distance of 3 m perpendicular to the side of the vehicle at a viewing height of 1.8 m above ground.
- 9.10.B.2** Headquarters must approve the use of abbreviations or initials. The Institution's name shall be larger and more prominent than any team sponsor name/logo. Additional graphics related to a team's institution(s) or sponsors are permitted, provided they are neither offensive nor disruptive.

9.10.C Event Logo

- 9.10.C.1** The Event logo shall be clearly displayed on both sides of the solar car and clearly visible from a distance of 3 m perpendicular to the side of the vehicle at a viewing height of 1.8 m above ground.
- 9.10.C.2** The logo will be provided by Headquarters and will measure no more than 200 mm in height by 500 mm in width.

9.10.D National Flag

- 9.10.D.1** The national flag of the country of the team must be displayed on both sides of the solar car adjacent to the windscreen and clearly visible from a distance of 3 m perpendicular to the side of the vehicle at a viewing height of 1.8 m above ground.
- 9.10.D.2** The minimum size of the flag is 70 mm by 40 mm.

9.10.E Front Signage

- 9.10.E.1** Solar cars must have an unbroken front signage area on the solar car body, forward of the windscreen. The entire front signage area must be visible in top view and in front elevation view and must not overlap the solar collector.
- 9.10.E.2** A 150 mm x 150 mm Event logo must be placed within the front signage area. The logo will be provided by Headquarters.
- 9.10.E.3** The front signage area shall also include the name of the team's host Institution(s).

10 Mechanical

10.1 Body Panels

All moving or removable body panels and the solar collector must be securely fastened to prevent unintended movement.

10.1.A Covers and Shields

All moving parts must be suitably covered to prevent accidental human contact when the solar car is fully assembled. The driver must be shielded from contact with all steering linkages and other moving parts.

10.1.B Clearance

Interference or rubbing of the wheels with the solar car's body, wheel well, or structure at full steering lock or suspension travel is not permitted. Movement of rod-end bearings must not be obstructed in any axis throughout the full travel of suspension and steering. Other moving parts, such as the motor shaft, must not contact stationary parts except through properly designed bearings. All wheels and their suspensions, steering linkages and geometries will be inspected for safe operation in normal and adverse conditions.

10.1.C Solar Collector Attachment

Teams shall provide two independent and different means of securing the solar collector (top shell of the car) to the chassis. This requirement is meant to provide additional reliability. The two methods could be for example 1) perimeter locking and 2) back-up lanyard.

If lanyards are used as one of the methods, the lanyards and the hardpoints on both the solar collector and chassis shall be capable of holding a 900 N (202 lbf) load. The lanyards shall not allow more than 600mm of upward travel of the solar collector. The forward hardpoint in both the solar collector and chassis shall be within 300mm of the forwardmost point of the seam that splits the collector portion from the chassis. The second hardpoint shall be behind the driver and on the opposite side of the vehicle centerline from the forward hardpoint.

Zip ties or tape are not allowed methods of solar collector attachment. Tape is allowed for aerodynamic sealing but shall not count as an attachment method.

10.2 Tire and Wheel Requirements

10.2.A Wheel Configuration

10.2.A.1 All vehicles shall have at least three (3) wheels. All wheel tires shall be in contact with the ground at all times.

10.2.A.2 The distance between the left-most wheel contact patch and the right-most wheel contact patch shall not be less than half the width of the solar car (at its widest point).

10.2.B Design Intent

The wheels and tires shall be designed for the intended application and able to withstand the loads and forces imparted by the vehicle's mass, speed capability, turning forces, and braking potential. Each wheel and tire on a single axle shall be rated for the full weight applied to that axle.

10.2.C Static Stability

The fully laden solar car shall not tip when tilted 45° about each pair of adjacent tire contact patches.

10.2.D Tires

- 10.2.D.1 Solar cars shall be fitted with tires that are compliant with US DOT standards, UNECE Regulation 30 or UNECE Regulation 75, as indicated by a DOT or E approval marking on the tire.
- 10.2.D.2 Tire inflation and loading must be within the manufacturer's rating at all times during vehicle operation.
- 10.2.D.3 If the tire is deemed to be a tube-type tire as per the manufacturer's specification, the appropriate tire tubes shall be used.
- 10.2.D.4 The speed rating of the tires shall be more than the maximum speed of the solar car. If not, the solar car will be speed limited to the rating of the tire.
- 10.2.D.5 The load rating of the tire shall be more than the maximum static load imposed by the fully-laden solar car.
- 10.2.D.6 The tires shall be free of any apparent defect.

10.2.E Wheels/Rims

The rim profile shall be shown that it is in accordance with (or matches) the bead requirements of the tire as specified by the tire manufacturer.

10.3 Cockpit

The cockpit must not subject the solar car occupants to excessive strain during normal operation, and must be designed to protect the occupants from injury in the event of an accident. The occupants must be provided adequate space for safe operation of the vehicle. Care needs to be taken in the design and construction of the vehicle to minimize the risk that any shafts or sharp objects could penetrate the cockpit in the event of a crash and potentially injure the occupants.

10.3.A Occupant Cell

- 10.3.A.1 **Roll Cage:** is the structural cage that encompasses the occupants from the level of the top of the shoulders upward. Any Occupant Cell structure above the shoulder is considered to be part of the roll cage.
- 10.3.A.2 **Structural Chassis:** is the tubular frame / monocoque composite chassis / hybrid of composite & tubular frame which encompasses the occupant's bodies, and to which the vehicle suspension system is connected.
- 10.3.A.3 All solar cars must be equipped with a roll cage that is fixed and integrally connected to the structural chassis.
- 10.3.A.4 The roll cage shall be constructed with metal elements. Composite roll cages are not allowed.
- 10.3.A.5 The portion of the structural chassis surrounding the driver below the roll cage shall be referred to as the Lower Occupant Cell.
- 10.3.A.6 The combination of the solar car Lower Occupant Cell and Roll Cage comprises the Occupant Cell.
- 10.3.A.7 Teams must provide documentation that specifies which parts of their solar car constitute the Occupant Cell.
- 10.3.A.8 The Occupant Cell shall encompass the occupant in all directions. When occupants are seated normally, with safety-belts and helmets on, no part of any occupant, nor the full free range of motion of the occupant's head (including helmet), may pass through planes drawn between the outer edges of adjacent tubes of the Roll Cage or Lower Occupant Cell, OR may intersect a line drawn between any two points on the outside-facing edge of adjacent tubes of the Roll Cage or Lower Occupant Cell.

Imagine stretching a rubber skin around the Roll Cage in a manner that it is in contact with the full length of every external tube; no part of any occupant or their helmet may touch the skin when fully buckled into the seat.

- 10.3.A.9 Each team must provide calculations, certified by the team's certifying reviewer, to show that the Roll Cage will not yield and all other components of the Occupant Cell will not deform by more than 25 mm and will not fail (exceed ultimate strength) at any point when subjected to the load cases outlined in Appendix E, section E.3.3, where g is the total gross mass of the vehicle including all occupants and ballast as submitted in the Mechanical PVDR Form (Appendix D).

- 10.3.A.10** The protection provided for the occupants in a collision must be documented in the team's Mechanical Vehicle Design Report as per Reg 5.3.A.
- 10.3.A.11** A preliminary sketch and description of the Occupant Cell must be submitted in the Preliminary Vehicle Design Report by the date in Reg 4.2.A.1, as per Reg 5.2.A.
- 10.3.A.12** In addition to providing collision and rollover protection, the roll cage must be designed so as to deflect body/solar collector panels of the car up and away from the occupants in the event of an accident. The front elements of the roll cage must be angled at least 15 degrees backwards from the vertical to facilitate deflection of the body/solar collector panel.
- 10.3.A.13** Wherever the Occupant Cell may come in contact with an occupant's helmet, the roll cage or structure must be padded with energy-absorbing material meeting SFI-45.1 or FIA 8857-2001 Type A or B, or better. This material must be bonded and secured to the structure, wrapping around 50% of the roll cage member or piece of vehicle structure.
- 10.3.A.14** A head restraint of at least 19 mm thick resilient material must be securely mounted behind the occupant's head without the use of cable ties, fabric straps, or temporary attachments. The headrest must support the occupant's head in normal driving position.

See additional regs interpretation in Appendix J

- 10.3.A.15** There must be 50 mm of clearance in all directions between any member of the Occupant Cell and the helmets of the occupants seated in the normal driving position. There must be at least 30 mm of clearance between the occupant's helmet and the padding to allow for free movement of the occupant's head.
- 10.3.A.16** Any carbon fiber panels rigidly attached to the Occupant Cell within 500 mm of the center of an occupant's head in a normal seated position and above the top of the occupant's shoulders shall have shatter resistant fabric (such as Kevlar or Dyneema) applied to the interior surface of the panel. The layer or layers shall total at least 5 oz/yd² of fabric weight.

In this context, "rigidly attached" includes any panel that is part of the Occupant Cell structure, or bolted or bonded to the Occupant Cell. This does not include panels that are part of a removable top shell held on by a solar collector attachment system as described in Reg 10.1.C. This regulation in no way allows for composite roll cages. It addresses composites that are not part of the Occupant Cell structure, but near the occupant's head or neck, as well as Occupant Cell panels below the shoulder, but near the driver's head or neck. The protection layers should have the minimum feasible number of cuts or breaks needed to conform to surface curvature.

10.3.B Occupant Seats

- 10.3.B.1** Single-Occupant solar cars shall be designed for 1 occupant with only one seat.
- 10.3.B.2** Multi-Occupant solar cars shall be designed for 2 to 4 occupants with a seat for each occupant.
- 10.3.B.3** Each solar car occupant must have a seat that faces forward at an angle less than 10 degrees of yaw from the forward direction of travel.
- 10.3.B.4** Each seat must have a back and a head restraint per Reg 10.3.A.14. The distance from the hip point to the top of the head restraint must be at least 800 mm for front seats and those of a single-occupant solar car and at least 750 mm for rear seats. (49 CFR 571.202a - Standard No. 202a; Head restraints). The hip point may be approximated as shown in the diagram below. Any additional seat padding must be included in this measurement.

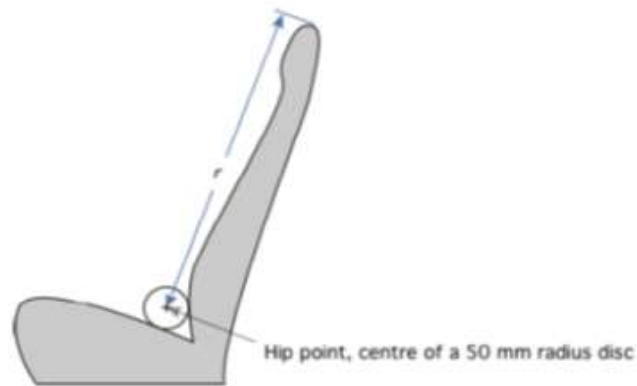


Figure 2 Hip Point Definition

- 10.3.B.5** Each occupant's heels must be below their hip point.
- 10.3.B.6** The angle between each occupant's shoulders, hips and knees must be more than 90 degrees.
- 10.3.B.7** Any additional seat padding must be positively secured to the seat.
- 10.3.B.8** Seats must be fixed in place fore/aft and not on adjustable rails. Consider accommodating drivers of different heights with adjustable pedals.

10.3.C Occupant Space

- 10.3.C.1** The occupant space for each occupant's upper torso shall be defined by an arc defined with an 835 mm radius measured from the hip point as defined in Appendix B of the occupant and projects forward 45 degrees from vertical, 25 degrees rearwards and 7 degrees side-to-side from the centerline of the occupant⁸.
- 10.3.C.2** The solar car structure, including the windshield must lie wholly outside the occupant space. The steering wheel, mirrors, seat backs, and head restraints may be inside the occupant space but must be designed to minimize the risk of injury to the occupant.
- 10.3.C.3** The driver's head must be above and behind the driver's feet. The seat must be appropriately constructed with a solid base and back rest.

10.3.D Belly Pan

The cockpit must be equipped with a full belly pan to isolate the occupants from the road - road debris must not be able to enter the cockpit, and the full free motion of the occupant limbs must be unable to come in contact with the road. The belly pan must be strong enough to support the full weight of each occupant. Each occupant's torso and limbs must be above the lower element of the structural chassis.

⁸ This minimum occupant space requirement is based on a 50-percentile male and does not allow for a helmet. Taller team members may need more occupant space.

10.3.E Safety Belts

- 10.3.E.1** All solar cars must be equipped with a minimum of a 5-point lap and shoulder belt harness system for each occupant.
- 10.3.E.2** The use of safety belts is mandatory.
- 10.3.E.3** The safety belts must be installed and attached securely to the structural chassis, as recommended by the manufacturer. Safety belt mounts should be designed to resist the same impact loads that the Occupant Cell is designed for (Reg 10.3.A.9).
- 10.3.E.4** If the belt passes through the seat, it must pass through without wrinkling, crimping or bending the belt excessively. All sharp edges shall be removed or covered to prevent cutting or fraying of the belt.
- 10.3.E.5** Only safety belt systems manufactured and certified to FIA 8853/98, FIA 8853-2016, SFI 16.1, SFI 16.5, or SFI 16.6 are allowed. Any modifications must be approved by the manufacturer. Certifications must be valid (not expired) for the duration of the event.
- 10.3.E.6** The placement of the attachment points for the seat belt harness shall be as follows (unless otherwise specified by the manufacturer):
- 10.3.E.7** The shoulder straps attachment point shall be rearwards between horizontal and highest of 30 degrees below horizontal and perpendicular to the occupant's spine or seat back.

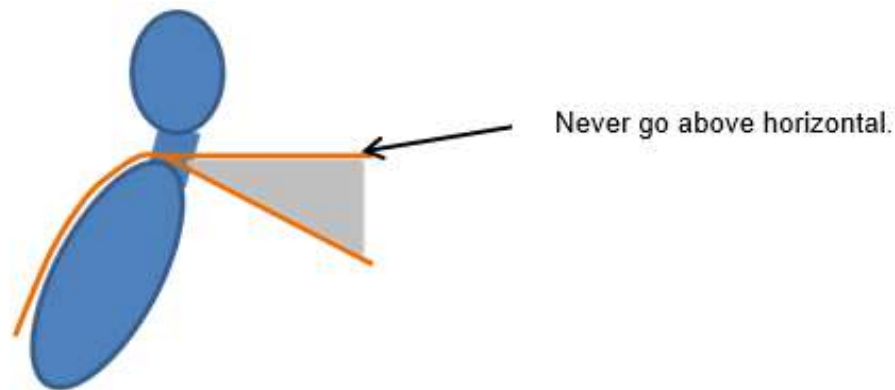


Figure 10.2 Range of shoulder strap attachment position

- 10.3.E.8** The shoulder belts shall be spaced wide enough apart to not squeeze on the neck, but narrowly enough that they will not fall off the shoulders. The mounting points shall extend backward and go inward by approximately one unit for every two units that the mounting point is located behind the point that the belt leaves the shoulder.

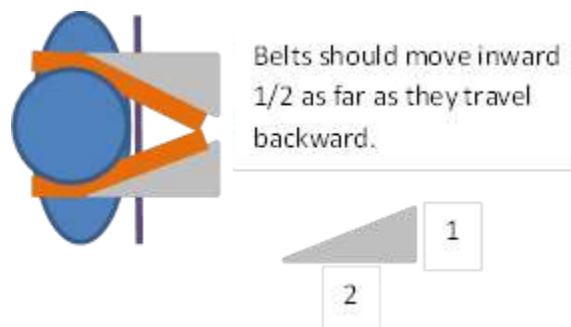


Figure 10.3 Horizontal configuration of shoulder strap attachment

- 10.3.E.9** The lap belt attachment point shall be downwards and rearwards from the occupant's lap between 60 degrees and 80 degrees from horizontal. The ends of the belt need to be well below the lap of the driver.

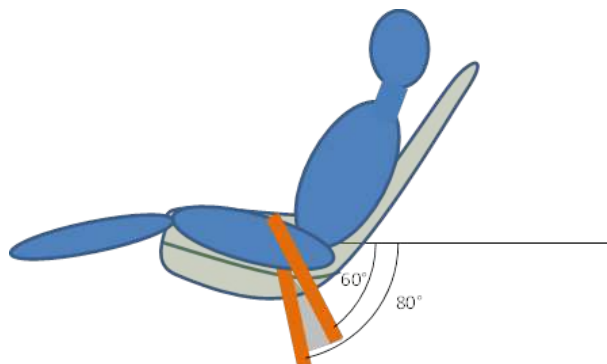


Figure 10.4 Configuration of lap belt attachment

- 10.3.E.10** The anti-submarine belt attachment shall be approximately 10 degrees forward of the plane of shoulder belts for 5-point or approximately 20 degrees rearward of the plane of shoulder belts for 6-point belts.

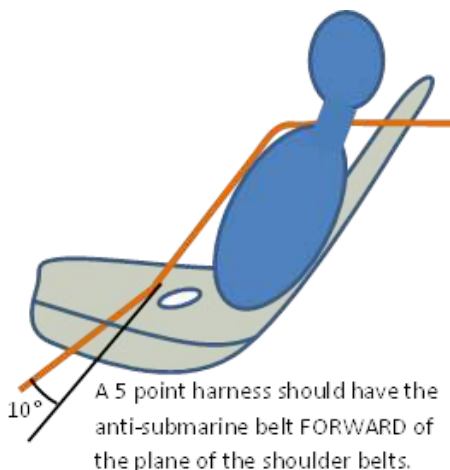


Figure 10.5a Configuration of anti-submarine belt attachment (5-point harness)

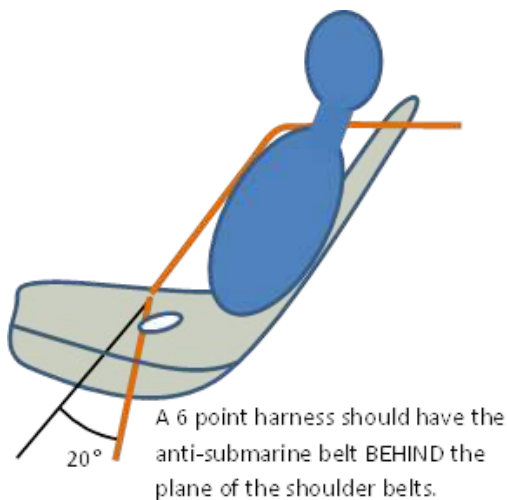


Figure 10.5b Configuration of anti-submarine belt attachment (6-point harness)

10.4 Fasteners⁹

All fasteners must be of suitable type, strength, and durability for their application. Friction, glued, or press fit assemblies will not be accepted in critical areas as the sole means of retention. For glued or press fit assemblies, a pin is required. The pin diameter shall be $\frac{1}{4}$ of the tube's outer diameter. A press fit roll pin is acceptable for this application. Set screws intended to transmit torque or force will not be accepted. Fasteners must meet the following minimum requirements:

10.4.A Bolts

Bolts used in critical areas must at minimum meet SAE grade 5, metric grade M8.8 and/or AN/MS specifications. Bolts must be of the correct length, and extend at least two threads beyond the nut. Bolts in tension must not have shaved or cut heads. All fasteners shall be properly torqued. U-bolts are not allowed in critical areas.

10.4.B Securing of Fasteners

All structural and other critical fasteners (bolts, nuts) must have an acceptable form of securing such that the fastener cannot loosen or be removed unintentionally. Acceptable methods of securing are:

- (1) Bolts with flex-loc or fujilok type nuts or other nuts that use flexure as the means of locking and are reusable.
- (2) Bolts with pre-drilled shafts and castle nuts with cotter pins installed to prevent loosening
- (3) Bolts with pre-drilled heads and/or nuts properly safety wired with stainless steel wire from 0.024" (0.6 mm) to 0.032" (0.8 mm) diameter conforming to Mil Spec MS20995C. The safety wire between fasteners and anchor points may not be used in shear and must be twisted to prevent loosening rotation of the fastener.
- (4) In blind hole applications, bolts with pre-drilled heads, properly safety wired.
- (5) Other methods of securing fasteners may be deemed acceptable at the discretion of the Inspector.

Securing methods that are not acceptable are Nylon lock nuts, Loctite, lock nuts that use thread distortion as a means to secure the nut, Nord-lock washers, or other "lock" washers. Lock nuts with thread distortion are not considered to be re-usable. Other methods of securing fasteners where the above methods are not appropriate may be considered at the discretion of the Inspector. Non-critical fasteners need not be secured with lock nuts.

10.4.C Securing Rod Ends

All rod ends shall be secured with jam nuts tightened with sufficient torque to prevent rotation. The jam nuts on rod ends do not need to be safety wired and do not need to be of the flex-loc type.

10.4.D Buckles and Straps

Plastic luggage type buckles or single push release straps are not considered acceptable means of securing any Critical Area. If nylon type straps are used in securing any Critical Area, ratchet type straps (without hook terminators) shall be used.

10.4.E Critical Areas

For application of the above critical areas are defined to include: steering, braking, suspension, seat mounts, safety harness, drive train, battery enclosure, ballast carrier, and parking brake.

10.4.E.1 Brake caliper systems present unique challenges to the securing of fasteners. Teams shall contact the inspectors prior to the event with any unique securing situations. Caliper securing will be evaluated in terms of total system redundancy.

10.4.F Hub Nuts

For designs using a single central hub nut to secure the wheel to the rotating assembly (hub, rotating axle, motor rotor, etc.), the threaded portion of the hub nut shall be at least 10.9 mm thick.

⁹ Fasteners are a complex subject and much care should be taken to choose appropriate fasteners for each application. Excellent references on proper use and securing of bolts can be found:

- Smith, Carrol. Prepare to Win.
- Reithmaier, Larry. Standard Aircraft Handbook.
- Federal Aviation Administration. FAA Advisory Circular AC43-13-1B. Chapter 7 (securing of fasteners)
- FAA-H-8083-30A. Aviation Maintenance Technician Handbook – General. pages 7-76 to 7-81

10.5 Brakes¹⁰

10.5.A Configuration

Solar cars must have a dual, balanced braking system so that if one system should fail, the solar car can still be stopped. The two systems must be operationally independent and must operate from a single foot pedal. The braking system can be front/rear or redundant front. Left/right redundancy is not permitted. Hydraulic systems must have separate master cylinders. Regenerative braking is not eligible as a redundant system.

10.5.B Brake Pads

Each brake pad used in the braking systems must have a contact area with the brake disk that is greater than 6.0 cm², and the pad must have full contact with the brake rotor. Pads must initially be at least 6 mm thick including the backing plate when installed on the car.

10.5.C Braking Performance

Solar cars must be able to repeatedly stop from speeds of 30 mph (48 km/h) or greater, with an average deceleration, on level wetted pavement, exceeding 4.72 m/s². Performance shall be demonstrated with mechanical braking only (no regenerative braking will be allowed during the test).

10.5.D Brake Lines/Cables

The brake lines (hydraulic or cable) shall be appropriately sized and constructed such that they have significant capacity beyond the pressure and loads that will occur under worst-case driving conditions.

10.5.E Brake Pedal Position

Left foot brake pedal actuation is allowed. If the team elects to have both accelerator and brake pedals operated by the right foot, the team must demonstrate the pedal arrangement allows for the foot to rapidly move between pedals. Refer to Reg 8.9.B for placement of the accelerator pedal if equipped.

10.5.F Hand Activated Brakes

Hand activated brakes are permissible if the driver can turn the steering wheel lock-to-lock without removing or repositioning either hand from the steering wheel.

10.5.G Cars with Mechanical (Hydraulic or Cable-actuated) Rear Brakes

10.5.G.1 For solar cars without anti-lock brakes, the front wheels must lock-up before the rear wheels.

10.5.G.2 Performance: Cars with mechanical rear brakes as one of their primary brake systems shall be able to demonstrate that the rear brakes can hold the car in place (front wheels elevated off the ground) on dry pavement under a forward pull equal to 15% of the car's weight in Event driving configuration with properly ballasted occupant(s) in place.

10.5.G.3 Volume Limiting Valve Systems: solar cars with mechanical rear brakes equipped with proportioning valve(s) require a means to lock-out the proportioning valve setting(s). The proportioning valve(s) shall be positioned out of any occupant's reach.

¹⁰ Braking systems for solar cars should be designed in accordance with standard automotive engineering practice. In general, bicycle type brakes are deemed to be too fragile for this kind of application and will not pass Scrutineering. This includes mountain bike type disc brakes. While such brakes might initially have enough stopping power to slow down a solar car, they do not have appropriate energy dissipation capability for long periods of brake application descending hills that are often common in the ASC Tour. Larger pads with more contact area can help the braking system dissipate sufficient heat to prevent failures involving boiling brake fluid, disc warpage, and loss of braking force. Additionally, bicycle disc brake pads often have very thin pad material. While sufficient to last for a few thousand miles on a bicycle, solar car teams in the past have worn entirely through them in well under a day of combined city/highway driving. Vehicles that do not heed these recommendations likely won't pass Scrutineering or be allowed to compete.

10.6 Parking Brake

Solar cars must be equipped with a parking brake.

10.6.A Performance

The parking brake shall be able to hold the car in place without wheel chocks on dry pavement under either a forward or rearward force equal to 10% of the car's weight in fully loaded condition.

10.6.B Independence

The parking brake must operate completely independently from the main braking system and must not be used in the performance tests specified in Reg 10.9.D.

10.6.C Locking

10.6.C.1 The parking brake must be able to be locked into the "ON" position, such that the driver does not have to continue to hold it to maintain position.

10.6.C.2 The driver shall be able to set and lock the parking brake while seated in the normal driving position and seat belted in.

10.6.C.3 The driver shall be able to set and lock the parking brake in a single action or motion.

10.6.C.4 The parking brake shall not disengage without explicit driver input.

Electrically operated parking brakes shall not disengage due to electrical faults such as loss of power.

10.6.D Contact Style

The parking brake shall not be of a tire or wheel contact style (i.e. pad on tire or pad on rim styles are not considered as acceptable designs).

10.7 Steering

10.7.A Steering Wheel

All steering in the vehicle must be controlled by the driver with a steering wheel designed to have a continuous perimeter as outlined in Appendix A. The steering wheel must be sufficiently strong to withstand loads the driver may impose on it.

Steering wheels 3D printed on hobby-grade FDM printers are extremely unlikely to pass inspection.

10.7.B Steering Stops

The steering system must include steering stops to prevent dangerous or damaging steering travel. Steering stops cannot be held in place by friction. They must be welded, pinned or bolted in place, and placed in the steering system in a way that will not create loads that will deform or scar the contacting pieces.

10.7.C Turning Radius

Solar cars must be able to make a U-turn in either direction, without backing up, such that any portion of the tire contact patches remain within a 15 m wide lane.

10.7.D Steering Backlash

The steering system must be designed with sufficient strength/stiffness and have minimal backlash so that the driver can always maintain good directional control of the vehicle.

10.8 Towing Hardpoint

Solar cars must be equipped with a hardpoint where an appropriate rope or strap may be attached in order to tow the car for emergency recovery purposes. The hardpoint must be either securely attached to or part of a non-moving structural component such that the car can be towed in the forward direction. The hardpoint or access to the hardpoint may be covered while not in use. The hardpoint must allow the car to be pulled with the body on the car; however, the canopy may be removed.

10.9 Dynamic Stability

Solar cars will be tested for dynamic stability and handling performance. A combination of the following tests may be conducted:

10.9.A Figure-8

Solar cars must be able to negotiate a Figure-8 course in less than 8 seconds per side. The Figure-8 course shall have a 5 m wide lane around two (2) 6 m radius center circles, as illustrated in Figure 3. The vehicle shall not knock over any of the cones or exhibit signs of structural instability. Clearance between moving parts of the car (eg, wheels) and stationary parts (eg, bodywork) shall be maintained per Reg 10.1.B.

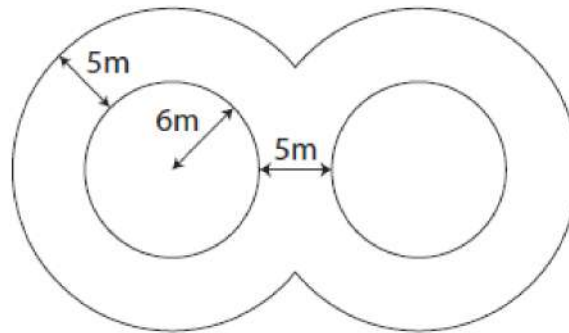


Figure 3 Figure-8 Course Layout

10.9.B Stability at Speed

10.9.B.1 Solar cars must be able to stay within a 3.5 m lane for at least 250 m. Cars must be able to achieve this regardless of crosswinds or gusting conditions. If a car cannot do this at 65 mph (104.6 km/h), the car speed will be limited to where it can stay within a 3.5 m lane for the entire Event.

10.9.B.2 If a team is observed to deviate from a lane three times during the Event, the maximum car speed will be limited to 10 mph (16 km/h) less than the recorded speed of the second lowest deviation, with a lower limit of 35 mph (56 km/h).

Examples: If a team is observed to deviate from their lane at 51, 55, and 59 mph, their speed will be limited to 45 mph. If a team deviates at 41, 41, and 45 mph, they will be limited to 35 mph.

10.9.B.3 If a team has had their speed limited per Reg 10.9.B.2 and is observed to deviate from the lane again, their speed limit shall be limited to 10 mph (16 km/h) less than the recorded speed of the deviation with a lower limit of 35 mph (56 km/h).

10.9.B.4 If a team has had their speed limited to 35 mph (56 km/h) and is observed to deviate from the lane, the car shall be disqualified until modifications are made to decrease instability and the vehicle is re-inspected by the Inspectors.

10.9.C Slalom Test

Solar cars must be able to negotiate a slalom course within 11.5 seconds. The slalom course shall be 126 m long, with cones equally spaced every 18 m as in Figure 4.

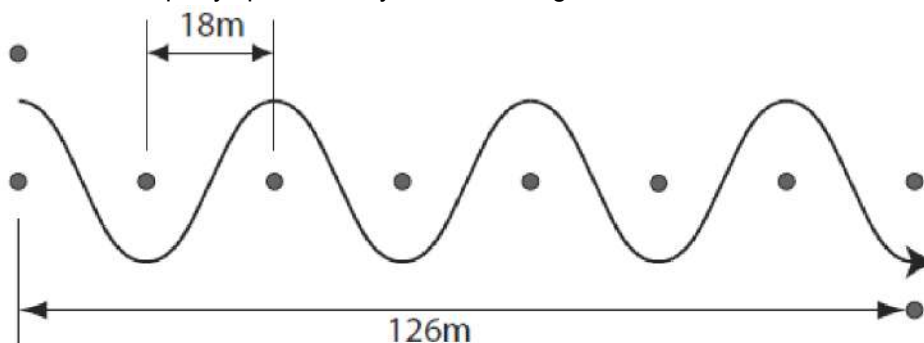


Figure 4 Slalom Course Layout

10.9.D Brake Test

Solar cars will be tested to verify compliance with Reg 10.5.C. The time interval over which the deceleration is averaged shall be from the first indication that the driver should stop until the solar car comes to a complete halt. When braking, the solar car must not veer excessively to the left or right, or exhibit structural instability. The tire pressure and mechanical systems settings used in this test will be considered as the driving configuration.

10.9.D.1 Solar cars may be required to demonstrate the brake performance a minimum of two (2) out of three (3) times.

10.9.E Acceleration Test

Solar cars will be required to accelerate from a standing start and time will be recorded at a distance of 18m from the start line. Cars must complete this distance in 8 seconds or less in order to be considered safe to drive in public traffic on public roads.. This test is a proxy for vehicle torque/hill climbing ability.

10.9.F Disqualification of a Driver

If it becomes apparent to the Officials that the solar car is capable of passing the required dynamic stability requirements, but a driver is not able to proficiently handle the solar car during one or more of the required components, the solar car driver may be disqualified at the discretion of the Officials.

11 Occupant Requirements

11.1 Occupant Registration

All solar car occupants (drivers and passengers) must be registered at Scrutineering. All occupants must be 18 years old or older.

11.1.A Drivers

11.1.A.1 Only registered solar car drivers will be allowed to drive solar cars during the Event.

11.1.A.2 Each team shall have a minimum of two (2) drivers available at all times and may register at most four (4) drivers.

11.1.A.3 All drivers must have a valid, non-expired driver's license. A copy of their driver's license must be submitted with the team Roster before Scrutineering (See Reg 4.4.B)

11.1.A.4 The solar car driver must be in the solar car during operation of the car.

11.1.B Passengers

11.1.B.1 Drivers are eligible as solar car passengers. Teams may register at most eight (8) additional solar car passengers.

11.1.B.2 The maximum number of occupants in a Multi-Occupant solar car shall be equal to the number of seats and seat belts provided.

11.2 Occupant Attire

11.2.A Clothing

Solar car occupants, while within the solar car, are exempted from the team uniform requirements from Reg 14.16. Clothing worn by solar car occupants must provide suitable cover and be non-offensive.

11.2.B Helmet

All solar car occupants must wear a helmet while the solar car is in motion. The helmet must meet or exceed the Snell (2015, 2020, or 2025 for Motorcycling, Karting, or Competitive Autosports), FIA (8859-2015 or 8859-2024), DOT FMVSS 218, ECE (22.05 or 22.06), AS/NZS 1698 standards, and will be inspected during Scrutineering.

An important note on DOT FMVSS 218 helmets: Unlike other helmet rating systems, the DOT does not actually certify helmets. Manufacturers are allowed to "self certify" that they meet the DOT requirements, and the internet is flooded with products bearing DOT markings that clearly do not actually meet DOT requirements.

As such, helmets that are only DOT FMVSS 218 and have padding less than $\frac{3}{4}$ " thick will not be allowed:

https://www.nhtsa.gov/sites/nhtsa.gov/files/documents/14283-identify_unsafe_motorcycle_helmets_070919_v4_tag.pdf

If the situation with DOT helmets does not improve, expect that helmets bearing only DOT ratings will not be allowed in 2028.

11.2.C Shoes

All solar car occupants must wear closed-toe shoes with a solid sole that will protect the occupant from debris that may be found on the road in the event of an accident in the solar car. Shoes that are securely fastened to an occupant's foot are preferred to avoid unintentional removal when driving or egressing the solar car. Occupant's shoes shall be approved at the Inspectors discretion.

11.3 Duration in Solar Car

Each occupant must not be in the solar car more than a cumulative total of six (6) hours in a given FSGP or ASC day. If the solar car is stopped on the side of the road such that the occupant is allowed to exit the vehicle for an extended period of time, this stopped time is not counted towards the six (6) hours driving time.

This applies regardless of role in the solar car. For instance, if an occupant drives the solar car for five (5) hours in a day, they may only remain in the solar car as a passenger for one (1) additional hour during that day.

11.4 Water/Fluids

Each occupant must have sufficient quantities of water/fluids in the cockpit area to stay properly hydrated. A minimum of two liters for each occupant must be provided. The water container must be secured in the solar car and the driver must be able to consume water while keeping at least one hand on the steering wheel.

11.5 Driver Communications

The solar car driver must have access to verbal and hands-free communication at all times with another team member outside the vehicle who is monitoring the solar car. Hands-free operation is defined as operation where the driver can activate the radio without removing their hands from the steering wheel.

If voice communication systems utilize volume detection rather than a push-to-talk button to initiate transmission, the communication system must be full-duplex.

11.5.A Cell Phone Use

Cell phones may not be used as the primary communication method with the solar car driver. Cell phones are permitted within the solar car as a backup communications method. Use of cell phones must comply with all local laws pertaining to cell phone use within a vehicle.

SECTION 3 – OPERATIONS

12 Tour

12.1 Format

The Tour consists of a series of Stages between predetermined locations (Stage Points) following a specific Tour Route. Stages have a mandatory Base Route and may include one or more Checkpoints and optional Loops. Each Stage will begin with a Stage Start, where all solar cars are released from the same point. At the end of any driving day during which an entry has not reached the next Stage Point (and does not Trailer to that Stage Point), the team will stop where they are along the Tour and begin the next morning at the same point. The Tour Stages are detailed in the Route Book and broken into Segments consisting of Base Legs (from one Stage/Check Point to the next Stage/Check Point) and optional Loops that may be completed to increase Event Distance Driven.

12.2 Unit of Distance

The mile is the unit of distance for the Event.

12.3 Determination of Winner

12.3.A Completion of Base Route

In each class, any team that fails to complete the Base Route of the Tour will automatically be ranked below all teams that successfully complete it. Loops, which are optional, are not considered part of the Base Route. Further class ranking is specified below.

12.3.B Single-Occupant Vehicles

Single-Occupant vehicle teams will be ranked based on Official Distance driven (higher distance is best). Should teams tie on Official Distance, Official Elapsed Time will be used as a tie-breaker to determine ranking (shorter elapsed time is best).

12.3.C Multi-Occupant Vehicles

Refer to Reg 13 for ranking of Multi-Occupant vehicle teams.

12.3.D Stage Winners

Stage winners for each class are determined by the highest ranked team using the criteria above.

12.4 Solar Car Configuration

Solar cars must drive in the same configuration as approved during Scrutineering and used at FSGP.

12.5 Support Vehicles

A solar car caravan consists of 3 vehicles: lead, solar, and chase. All vehicles/trailers associated with a team other than the solar car itself are considered support vehicles, must be registered with Headquarters, and must meet US/Canada Federal Motor Vehicle Safety Standards.

12.5.A Drivers

A minimum of two (2) drivers should be available for each vehicle associated with the team.

12.5.B Lead Vehicle

Each team must provide a caravan lead vehicle to alert oncoming traffic of the solar car's presence.

12.5.B.1 The lead vehicle must travel within 500 m (0.3 mi) ahead of the solar car, with its headlights on and with roof-mounted flashing amber lights visible from ahead (single bulb rotating beacon lights are not acceptable).

12.5.B.2 The lead vehicle must not tow a trailer.

12.5.B.3 The lead vehicle shall not be taller or longer than a standard 15-passenger, full-size van.

12.5.B.4 The lead vehicle shall carry at a minimum the following safety equipment: certified, stocked first aid kit, an ABC fire extinguisher (10 lb / 4.5 kg active material capacity or larger), safety vest (1 per occupant), four (4) orange cones (minimum 300 mm high), and an orange warning flag.

12.5.C Chase Vehicle

Each team must provide a caravan chase vehicle to protect the solar car from the rear.

- 12.5.C.1** The chase vehicle must follow directly behind the solar car; with roof-mounted, flashing amber lights visible from the front and the rear (single bulb rotating beacon lights are not acceptable).
- 12.5.C.2** The chase vehicle must not tow a trailer.
- 12.5.C.3** The chase vehicle shall not be taller or longer than a standard 15-passenger, full-size van.
- 12.5.C.4** The chase vehicle shall carry at a minimum the following safety equipment: certified, stocked first aid kit, ABC fire extinguishers (total 20 lb / 9.0 kg active material capacity or larger), safety vest (1 per occupant), four (4) orange cones (minimum 300 mm high), orange warning flag, battery MSDS, battery spill kit and method of containment of battery fires / 40 kg of sand.
- 12.5.C.5** The chase vehicle shall be equipped with a commercially available/unmodified GPS unit that displays the current vehicle speed and is visible to the Observer.

12.5.D Scout Vehicle

Each team is permitted to have a scout vehicle for the purpose of investigating road and traffic conditions ahead of the solar car.

- 12.5.D.1** The scout vehicle should maintain at least a 0.5 mi (0.8 km) separation from their solar car caravan and the solar car caravans of other teams.
- 12.5.D.2** The scout vehicle must not obstruct traffic or other solar car caravans.
- 12.5.D.3** The scout vehicle shall not be taller or longer than a standard 15-passenger, full-size van.

12.5.E Other Support Vehicles

Additional support vehicles, including truck/trailer units, may travel on the Tour Route, but should maintain at least a 0.5 mi (0.8 km) separation from all solar car caravans. These other support vehicles must not obstruct traffic or other solar car caravans.

12.5.F Support Vehicle Graphics

All support vehicles, including trailers, must be marked with:

- 12.5.F.1** Team number (at least 250 mm tall, with a 40 mm brush stroke) on both sides and the rear.
- 12.5.F.2** Team number (at least 150 mm tall) on the top passenger side of the front windshield.
- 12.5.F.3** Name of the team's host Institution(s) displayed prominently on both sides.
- 12.5.F.4** **Event Logo:** Headquarters will provide two (2) Event Logos per support vehicle to be placed on the sides of each vehicle and trailer. These logos will not be larger than 310 mm in height by 460 mm in width.
- 12.5.F.5** **Slow Moving Caravan:** A sign must appear on the rear of the chase vehicle to warn overtaking traffic of the solar car caravan. Signs will be provided by Headquarters at a size of no larger than 500 mm by 500 mm. Teams desiring to create their own signs may do so provided "CAUTION: SOLAR CAR CARAVAN AHEAD" is the wording used and is clearly indicated on a contrasting background. Alternative signage must be pre-approved before arrival at Scrutineering by the Inspectors.
- 12.5.F.6** Additional graphics are permitted, provided they are neither offensive nor disruptive.

12.6 Radios / Communication

The caravan chase vehicle must have access to two-way radio communication with the solar car driver at all times. All teams must also have a separately monitored CB radio in every support vehicle on the route tuned to an "official event" CB channel to communicate with other nearby teams and Officials.

12.6.A Observer Monitoring

All solar car driver communications must be audible (for voice communications), to the Observer at all times. Any communications in code or in languages other than English, must be explained to the Observer if requested.

12.7 Observers

Trained Observers selected and sponsored by Headquarters, will travel with each team to alert the Inspectors to possible infractions of these Regulations, and to help report unforeseen events to Headquarters. Observers are not authorized to interpret these Regulations or give advice on Tour strategy. Observers may be rotated in their team assignments at Checkpoints and Stage Points.

12.7.A Observer Access for Inspection

Observers will be assigned to keep each solar car in sight throughout the Tour unless the vehicle or battery enclosure(s) are impounded. Observers shall witness and note any and all work done on the solar cars. Observers must be allowed access to the solar cars for inspection of ballast during all driver changes.

12.7.B Observer Logbook

The details of the activities of a team will be recorded in a Logbook carried by the Observer. The team leader and advisor will be permitted to review and sign the Logbook each day; however, failure to do so does not make any record invalid. The records kept by the Observer include the Start Times, Stop Times (including Checkpoints and Stage Points), Impound Times, Load On Trailer Times, Route Steps successfully completed, and any apparent rule infractions either by their assigned team or by any other team.

12.7.C Observer Accommodations

12.7.C.1 During Tour Hours: Teams may designate the front passenger seat in the chase vehicle as the Observer's seat, otherwise they must allow the Observer to choose any seat in the chase vehicle's second row. The Observer must be able to see the solar car and read the chase vehicle's speed via a commercially available/unmodified GPS unit from this location. The Observer must also be able to determine, at least periodically, how many vehicles are following behind the team.

12.7.C.2 Before and After Tour Hours: Teams must provide a secure shelter for the Observer of equivalent quality to the best provided accommodations for any team member or advisor. Teams must also provide a secure location approved by the Observer for the battery impound box.

12.7.C.3 Meals and Lodging: Observers should be considered a guest of the team for whom the team will supply proper meals, drink, shelter, and amenities afforded to any member of the team or advisor.

12.8 Briefings

A Briefing will be held at each Stage Point. A special meeting may be called in cases of emergency. Attendance at meetings by a team representative and driver(s) is required. Briefing notes and other daily updates will be available at Stage/Check Points and/or communicated through each team's Generic Team Email. All official statements, rule interpretations, and special instructions will be contained in these postings. On days not ending at a Stage Point, it will be the responsibility of the team to check available outlets for updates and instructions.

12.9 Timing

12.9.A Responsibility

Headquarters will only recognize timing and distance information from the Timing Officials.

12.9.B Official Elapsed Time and Official Distance

At the conclusion of each Stage of the Tour, an Official Elapsed Time in h:mm:ss and an Official Distance in miles will be calculated for each entry. The summation of the Stage results will yield a total Official Elapsed Time and total Official Distance for the entry.

12.9.B.1 Official Distance = Event Distance Driven – Deductions

Event Distance Driven is the sum of Segment distances driven including all Base Leg Route Steps and valid Loops completed by the solar car. Deductions consist of Penalties and Protest Filing Fees which are assessed in Miles.

12.9.B.2 Official Elapsed Time = Sum of Elapsed Segment Times – Overnight Stop Times – Mandatory Hold Times – Trailing Durations

Elapsed Segment Times are calculated based on elapsed time between Official Start and Official Stop driving times during each Segment. For teams that Trailer, the Trailing Duration is from their solar car's Load On Trailer Time to their next Official Checkpoint Arrival Time or Official Stage Finish Time.

12.9.C Location Tracking

If location tracking devices are available, all teams must affix the unit provided by the Organizers in their solar car (typically under the windshield with a clear upward view) such that it reliably transmits real-time data throughout each Tour Day. Additional instruction will be provided at the Event.

12.9.D Team Off-Course

If a team departs from the Tour Route, but returns to that intersection or point and properly continues, their Elapsed Time will be determined in the normal manner; no credit will be given for the time the team was off-course.

12.10 Tour Hours

12.10.A Tour Day

12.10.A.1 Standard Tour Day: Typical Tour Day driving hours start at 9:00 am and stop at 6:00 pm.

12.10.A.2 Nominal Tour Day: The driving hours for a specific Tour Day, which may be adjusted later or earlier than the standard times based on the segment distances and coordination of activities at each of the Stage/Check Points. Any non-standard Tour Day timing will be announced in advance of the Tour.

12.10.A.3 Official Tour Day: The Official driving hours on a Tour Day for a specific team, including start and stop timing adjustments from Nominal per Regs 12.12.E, 12.13, and 12.15.A.

12.10.B Time Zones

The same official clock time (Tour Time) as displayed by the Tour Officials at the Stage Start Line will remain in effect for each team for the entire Stage, even though that Stage's route may cross through different time zones. The official time zones for each Stage will be announced in advance of the Tour.

12.11 Tour Route

12.11.A Route Book

The ASC Route Book will be distributed to each team that qualifies for the Tour. The Route Book will contain information to direct the team along the official route. It will specify days, distances, directions, Route Steps, maps, and points of reference. For a team to receive Official Distance or Time, they must follow the official Tour Route.

12.11.B Route Revisions

Due to unforeseen events, it may be necessary to detour from the official route. When advance warning is available, Tour Headquarters will correct the official route accordingly and provide revisions to the Route Book to all Tour teams, or provide written revisions at the Briefing or at Checkpoints.

12.11.C Teams Departing from the Tour Route

Any team leaving the Tour Route must rejoin the route at the same intersection or point where they left the route, or they will be considered to have trailed from their last completed route step before going off route. Their Load On Trailer Time will be the time that they went off route.

12.11.D Emergency Route Revisions

Due to unforeseen events, it may be necessary to detour from the official route (emergency road closures etc.). In the event that a team shall have to deviate from the official route due to an unplanned road closure that was not previously communicated to the team, the team shall:

- (1) contact Headquarters for further instructions;
- (2) follow the published detour and rejoin the official route at the earliest possible location; and,
- (3) document the route taken, the start and end times in the detour from the official route, and submit the route to Headquarters for evaluation.

12.12 Stage Starts

Teams are released from the Stage Start location in 1-minute intervals at the start of the Nominal Tour Day. Each team's Lead and Chase vehicles must merge with their solar car after it leaves the Start Line. The movement of all vehicles in the Start Line area will be controlled by Start Line Officials.

12.12.A Starting Order / Assigned Start Time

For the first day of the Tour, Assigned Start Times will be determined based on Official Total Laps from the FSGP Qualifier, ranked from highest to lowest. If there is a tie, the team with the fastest recorded FSGP lap time will start first. For all other Stage Starts, the Assigned Start Times are based on the Official Distance from the previous Stage, ranked from longest to shortest. If there is a tie, the team with the lowest Official Elapsed Time for that Stage will start first.

12.12.B Teams Not Ready / Start Release Time

Typically a team's Assigned Start Time is also their Start Release Time.

However, if a team's solar car, lead, and chase vehicles, with drivers in each vehicle, are not ready in their assigned starting positions 15 minutes before the start of the Nominal Tour Day, the Start Line Officials may, at their discretion, move the Start Release Time of all of the following cars up one slot, and the tardy team must move to the end of the starting queue.

If multiple teams are not ready on time, those teams will be released in the order that they arrive ready to start.

12.12.C Official Stage Start Time

Assigned Start Times will be distributed during that morning's charging session. If a team departs at or after the time they are released by Start Line Officials, then their Official Stage Start Time is the earlier of their Assigned Start Time or Start Release Time. If a team departs before their Start Release Time, then their Official Stage Start Time is their Actual Start Time and an early start penalty applies per Reg 15.5.I.

12.12.D Delayed Start

Any Stage Start may be delayed if inclement weather or other hazardous conditions appear likely to pose a threat to the solar cars or their drivers. If a Stage Start is delayed, then all Assigned Start Times for that day will be adjusted accordingly.

12.12.E Stage Start Timing Adjustment

The 1-minute start time interval at the beginning of each Stage will be accounted for at the Stage Finish. The Official Stage Close Time for each team will be adjusted from the Nominal Stage Close Time to reflect the offset from their Official Stage Start Time to the start of the Nominal Tour Day¹¹. Checkpoint Close Times, End of Day times, and impound times will not be adjusted by the Stage Start Timing Adjustment.

12.13 Overnight Stops¹²

On days not ending at a Stage Point, teams will end the day along the route at a location of their choosing. A Grace Period of up to 15 minutes early or 30 minutes late is applied to the Nominal End of Day time (6:00 pm unless otherwise announced) to allow time for teams to find a safe place to stop. It is possible that the overnight stop location could be in a Loop.

The Actual Overnight Stop Time and nearest Route Step where the team declares End of Day is to be recorded by the team's Observer. If a team declares End of Day within or after their Grace Period, their Official Overnight Stop Time will be their Actual Overnight Stop Time. If a team declares End of Day before the start of the Grace Period, their Official Overnight Stop Time will be set to 15 minutes before the Nominal End of Day time.

The offset from the team's Official Overnight Stop Time to the Nominal End of Day time will correspondingly be applied to the Nominal Morning Start Time (9:00 am unless otherwise announced) to determine the team's Corrected Start Time for the next morning.

The next morning, solar cars will resume driving from the same location where they declared End of Day the previous evening. The team's Actual Morning Start Time is to be recorded by the team's Observer.

If a team departs at or after their Corrected Start Time, their Official Morning Start Time is their Corrected Start Time. If a team departs before their Corrected Start Time, their Official Morning Start Time is their Actual Morning Start Time and an early start penalty applies per Reg 15.5.I.

Since Inspectors and Timing Officials will not necessarily be present, teams are responsible for managing Overnight Stops on their own, including when/where to declare End of Day, solar charging, determining their Corrected Start Time, and resuming driving from the proper location. Observers will record what the team does but do not have the authority to tell teams what they can and can not do, nor determine the proper times to stop and restart.

¹¹ Example: Team A has an Official Stage Start Time of 9:07 am; 7 minutes after the 9:00 am Nominal start time. Their Official Stage Close Time is adjusted to 6:07 pm; 7 minutes after the 6:00 pm Nominal close time.

¹² Examples: Consider standard 9:00 am - 6:00 pm Tour Days.

- Team B drives past the end of the day to find an appropriate overnight venue, and stops at 6:17 pm. This is their Official Overnight Stop Time. The next morning, Team B's Corrected Start Time will be 9:17 am.
- Team C finds a good place to stop for the night prior to the end of the day, and stops at 5:53 pm. This is their Official Overnight Stop Time. The next morning, Team C's Corrected Start Time is 8:53 am.
- Team D elects to stop driving at 5:40 pm (20 minutes earlier than the end of the day, greater than the 15 minute early Grace Period). Team D's Official Overnight Stop Time is 5:45 pm. The next morning, Team D's Corrected Start Time is 8:45 am. They have effectively forfeited 5 minutes of available driving time.
- Team E drives all the way until 6:35 pm (35 minutes later than the end of the day, greater than the 30 minute late Grace Period). Team E's Official Overnight Stop Time is 6:35 pm. The next morning, Team E's Corrected Start Time will be 9:35 am. They have not achieved any extra driving time, and they will collect a penalty for driving 5 minutes outside of authorized times per Reg 15.4.I.

12.14 Checkpoints

A Checkpoint is a mandatory stop in a predetermined location along the Tour Route. Checkpoints will remain open for a specified amount of time to be announced in advance of the Tour.

12.14.A Open Checkpoints

A team's Official Checkpoint Arrival Time is based on their entry at an open Checkpoint. Teams reaching an open Checkpoint will be subject to a mandatory Hold Time (typically 30 minutes). If a team Trailers to an open Checkpoint, they are exempt from the Checkpoint Hold Time and their Official Checkpoint Arrival Time will be the Driving Resumption Time (Reg 12.27.D). The most recent Route Step driven on solar power and the time that the solar car is loaded on the trailer will be recorded by the team's Observer. Trailering teams need to notify the Checkpoint Timing Officials that their solar car has entered the Checkpoint location. This time of notification will be logged as their Trailer Arrival Time.

Failure to stop at an open Checkpoint results in no credit for Event Distance Driven beyond that point. Movement of all team vehicles within the Checkpoint shall be under the control of Checkpoint Officials.

If a team finishes serving mandatory Hold Time at or after the Checkpoint Close Time, their Official Checkpoint Departure Time will be their Checkpoint Release Time. Otherwise, if a team doesn't have Hold Time after the Checkpoint closes but departs to proceed along the next Base Leg after the Checkpoint Close Time, then their Official Checkpoint Departure Time is the Checkpoint Close Time. In all other cases, their Official Checkpoint Departure Time is their Actual Departure Time. If a team departs before their Checkpoint Release Time, an early start penalty applies per Reg 15.4.I.

12.14.B Checkpoint Release Time

If a team runs Loops, their Checkpoint Release Time will be when they finish serving their Loop Hold Time after the successful completion of their final valid Loop. If a team doesn't run Loops, their Checkpoint Release Time will be when they finish serving their Checkpoint Hold Time. However, if the Checkpoint has a published Earliest Release Time, no teams are released to proceed along the next Base Leg before this time.

For teams that Trailer to an open Checkpoint and wish to continue driving the next Base Leg, their Checkpoint Release Time will be the Driving Resumption Time (Reg 12.27.D). If multiple solar cars are trying to leave the Checkpoint at the same time, priority will be given based on the earliest Official Checkpoint Arrival Time. For teams that Trailered to the checkpoint, departure priority will be based on the earliest logged Trailer Arrival Time.

12.14.C Closed Checkpoints

After the specified number of open hours, Checkpoints are closed. Teams are not required to stop at closed Checkpoints and should immediately proceed on the Tour Route.

12.14.D End of Day at Checkpoint

Any teams that have Hold Time after the Checkpoint closes will be required to finish serving the full duration of their Hold Time. If a team is at a Checkpoint when their end of day Grace Period starts, they must declare their End of Day at the Checkpoint location and begin their evening solar charging session. If a team has any remaining Hold Time after declaring End of Day, it will be served the next morning by incorporating the remaining Hold Time duration into their Corrected Start Time.

12.15 Stage Finishes

A Stage Finish is a mandatory stop for all teams to regroup in a predetermined Stage Point location along the Tour Route. The Stage Finish will open and close at predetermined times, to be announced in advance of the Tour.

A team's Official Stage Finish Time is based on their entry at the Stage Point. If a team fails to arrive at a Stage Point before their Official Stage Close Time or Trailers to the Stage Point, their Official Stage Finish Time will be their Official Stage Close Time.

Once a team's solar car crosses the Stage Finish Line, the movement of that team's vehicles shall be under the control of Finish Line Officials. Specific areas will be designated for solar charging, impound, support vehicle parking, and Tour Headquarters.

12.15.A Official Stage Close Time

Each team's Official Stage Close Time is calculated per Reg 12.12.E and may include additional adjustments from Timing Officials to ensure that all teams receive the same available Tour hours each day. All mandatory Hold Time expires after a team's Official Stage Close Time. Any team not at the Stage Point by their Official Stage Close Time will be considered to have Trailered. The most recent Route Step driven on solar power prior to the team's Official Stage Close Time and the time that the solar car is loaded on the trailer will be recorded by the team's Observer.

12.16 Loops

At select Stage/Check Points, teams will have the option to repeatedly drive optional Loops to increase their Event Distance Driven for that Segment. The Loops will be published in the Route Book.

12.16.A Loop Timing

12.16.A.1 At Checkpoints, teams may proceed onto the Loop after serving their Checkpoint Hold Time. At Stage Finishes, teams must serve a mandatory Hold Time (typically 30 minutes) before they begin Loops. Loops cannot be started in the End of Day Grace Period. For teams that complete at least one valid Loop, their Official Loop Start Time will be when they begin their first loop from the Stage/Check Point.

12.16.A.2 At the conclusion of each valid Loop, teams must serve a mandatory Loop Hold Time (typically 5 minutes) at the Stage/Check Point before they begin their next Loop (or at Checkpoints, elect to continue forward along the next Base Leg). If multiple solar cars are trying to start a Loop at the same time, priority will be given based on the earliest Official Checkpoint Arrival Time or Official Stage Finish Time. If the Checkpoint Loop Close Time passes while a team is in their Loop Hold Time they shall finish the Loop Hold Time before proceeding along the Route. A team's Official Loop Finish Time will be the actual finish time of the last valid Loop they successfully complete before the Loop/Stage Close Time.

12.16.A.3 Only distance for valid Loops will be credited. Loops that are only partially completed by the Loop Close Time (for Checkpoints) or by the team's Official Stage Close Time (for Stage Points) are not valid. Teams that fail to complete a Loop before the Checkpoint Loop Close Time will resume the Event from the Checkpoint. If a team begins a Loop before finishing serving their prior mandatory Hold Time or Trailers their solar car during the Loop, that Loop is invalidated.

12.16.A.4 If a Multi-Occupant vehicle team drops one or more occupants while driving a Loop, that will impact their minimum number of occupied seats during that entire Segment of the Tour.

12.16.B Multiple Loop Options

12.16.B.1 At select Stage/Check Points, teams may have multiple optional loop options to choose from. All options will be published in the Route Book. At these locations teams may elect to drive these loop options in any order and combination.

12.16.B.2 Teams must inform their observer which loop option is being attempted.

12.16.B.3 Teams may elect to change the loop option they are attempting during the attempt. Teams must inform their observer of this change immediately. If the solar car is past the location where multiple loop options diverge at the time of the change and wishes to backtrack to drive a different loop, the team will be deemed to have departed the route from the point of divergence in accordance with Reg 12.11.C and must resume driving the new loop choice in accordance with Reg 12.27.D.1.

12.16.B.4 Upon successful completion of a loop, the team and their observer must inform HQ of the loop option they have completed prior to departing for another loop attempt or continuing along the route.

12.16.B.5 When multiple loops are available at one location, some (but not all) loops may designate a higher minimum speed than Reg 12.22. This will be indicated in the Route Book.

12.17 Impound

12.17.A.1 Observers secure solar car batteries overnight in the impound configuration (Reg 8.11) that was inspected during Scrutineering. Teams should recognize that they will need additional power sources if they have to power or troubleshoot vehicle systems while batteries are impounded.

12.17.A.2 Headquarters shall be appraised of special issues for impound.

12.17.B Impound Times

12.17.B.1 Impound time starts at 8:00 pm each evening and ends at 7:00 am the following morning. Teams that fail to properly impound will be assessed the impound penalty as per Reg 15.5.M.

12.17.B.2 Impound is not required for MOV teams that elect to charge from unmetered external energy. Any time a team decides to charge from unmetered external energy, they must first officially declare this intention to their Observer and notify the Event Organizers as soon as possible.

12.18 Solar Car Mandatory Hold Period Activity

Solar charging of solar car batteries and solar car maintenance are allowed while serving mandatory Hold Time. Solar collector stands that are carried by the solar car are allowed to be used. MOVs are not permitted to perform external energy charging. Teams must not interfere with or block any other team's movement. Teams unable to leave after their mandatory Hold Time may be required to move their solar car elsewhere.

12.19 Solar Charging Areas

12.19.A Stage Point

A solar charging area will be provided for the teams. Internal combustion generators will not be permitted within the solar charging area. Solar car charging must occur within this designated area, and activities that may disrupt or interfere with charging must not take place in this designated area.

12.19.B Overnight Stop Between Stage Points

Teams are responsible for finding their own solar charging area on days not ending at a Stage Point.

12.20 Traffic Laws

During the Tour, all state and local traffic laws must be obeyed. (Note: while Event Organizers might not be aware of or enforce specific local regulations, under no circumstances does this imply that local jurisdictions will not enforce their own local ordinances, laws, or regulations.)

12.21 Maximum Speed

Solar cars must observe a maximum speed limit of 65 mph (104.6 km/h) or the posted speed limit (whichever is lower), unless further restricted by the Officials per Reg 10.9.B.

12.22 Minimum Speed

In areas with a posted speed limit of at least 60 mph (96.6 km/h), teams shall maintain a minimum speed of 20 mph (32.2 km/h) unless trapped behind other traffic. Teams must also observe posted minimum speeds. Teams unable to maintain this speed must pull over at the next safe opportunity to resolve the issue.

12.23 Passing Teams

In the event that one team is overtaken by another, the overtaking team must signal their intention to pass by attempting to make CB radio contact with the team being passed to coordinate the pass and flashing the headlights of their lead vehicle between high and low beam. Once the overtaking team has signaled their intention to pass, the team being passed must facilitate the pass at the first available safe opportunity, either by slowing down by at least 5 mph (8 km/h) in a zone where passing is permitted and feasible, or by pulling completely out of the traffic lane.

12.24 Allowing Traffic to Pass

When six (6) or more vehicles are lined up behind a team's chase vehicle, (including other teams' solar car caravans wishing to pass) the team must pull over as soon as safely possible to let traffic pass.

Teams need not disrupt their own progress to let other vehicles pass when they themselves are traveling at the posted speed limit or trapped behind other traffic.

12.25 Drafting

Drafting by a solar car is prohibited. A solar car will be considered to be drafting if it continuously follows behind another vehicle at less than a 3-second interval. The only exception to this is in congested traffic at speeds of 25 mph (40 km/h) or less.

12.26 Pushing

Except for the following situations, solar cars must not be pushed or pulled during the Tour Day. In no case shall regenerative braking be engaged while pushing or pulling the solar car.

12.26.A Stage/Check Points

Solar cars may be pushed within the confined area of the Stage/Check Point from when they enter by crossing over the timing line until they cross the timing line again to continue driving the Tour Route.

12.26.B Emergency / Breakdown

In an emergency or breakdown situation, the solar car must be removed from the road as quickly as is prudent. In this circumstance, the car may be pushed or lifted off the roadway. Upon resuming driving, the solar car may then be pushed or lifted back onto the roadway to the same location where it left the roadway.

Quick repairs may be done on the shoulder of the roadway, but in the case of major repairs, teams are encouraged to make use of this rule to relocate their car to a safer area to work away from traffic!

12.26.C Weather

The solar car may be pushed onto and off of a trailer to protect it from the weather, provided the solar car is moved back to its original location after it is unloaded from the trailer.

12.26.D Penalties

Any pushing outside of the circumstances outlined above that advances the solar car along the route will incur a penalty as per Reg 15.5.F.

12.27 Trailing

A team is considered to have Trailed if for any reason their solar car failed to successfully complete a Base Leg or a Loop.

12.27.A Pushing

Should it become necessary to load the solar car onto a trailer, it may be pushed onto the trailer.

12.27.B Battery Charging

Battery charging from the solar collector while Trailing is allowed during non-impound hours.

12.27.C Trailing to Stage/Check Points

If a Trailing team fails to reach the next Checkpoint along the Tour before it closes, they must continue Trailing to the next open Checkpoint or the Stage Finish. If any team fails to arrive at the Stage Point by battery impound time on the final day of the Stage, they will be assessed failure to impound penalties according to Reg 15.5.M. It is on the onus of the team to notify Stage/Check Point Timing Officials of their solar car's arrival by Trailer.

12.27.D Resumption of Driving

If a team has Trailed on a Segment, they may resume driving via one of the following options:

12.27.D.1 At the point they started Trailing, or earlier along the route (if successful they will no longer be considered to have Trailed from that point).

12.27.D.2 At an open Checkpoint. Teams using this option will be released to continue the Tour at the posted Driving Resumption Time for that Checkpoint. Should multiple teams be resuming driving from the same Checkpoint, teams will be released in the order of their logged Trailer Arrival Time.

12.27.D.3 At the start of the next Stage.

A team will not be eligible to start any additional loops on that Segment after Trailing.

12.28 Accommodations and Lodging

All teams are responsible for team accommodations and food throughout the Tour. Teams are responsible for their own reservations.

When overnight locations are known well ahead of time, such as Stage Points, we recommend that you reserve lodging well before the event!

12.29 Accidents and Re-Inspection

All accidents involving either solar cars or support vehicles must be reported immediately to Headquarters. In the case of an accident involving personal injury notification of the appropriate emergency medical services and public safety officials shall take priority. If a solar car is involved in an accident it must:

- (1) Stop and be visually inspected by team members and the Observer.
- (2) Be re-inspected by an Inspector at or before the next controlled stop (Stage/Check Point). The Inspector may require repairs prior to resuming the Tour.

12.30 Alterations and Repairs

If a team makes alterations or repairs to the solar car that may affect the safety, handling, or roadworthiness of the solar car, they must report these changes to Headquarters at the next Stage Point. The alterations/repairs must be inspected before the team is allowed to resume the Tour.

12.31 Team Uniforms

Reference Reg 14.16 for team uniform requirements during the Tour.

13 Multi-Occupant Vehicle Scoring

13.1 Goal

The goal for Multi-Occupant vehicles is to efficiently transport people as far as possible with considerations for vehicle practicality.

13.2 Scoring

Multi-Occupant vehicle teams will be ranked by the following formula, where the highest score is best:

$$\text{Score} = 40 \times (D/D') + 40 \times (O/O') + 20 \times (P/P')$$

Where D is the Official Distance credited, O is the Optimization Score, and P is the Practicality Score, and where D', O', and P' are the maximum values achieved by any MOV in the Event.

13.2.A Official Distance (D)

Official Distance is defined in Reg 12.9.B.

13.2.B Optimization Score (O)

The Optimization Score of an MOV will be calculated as:

$$O = (AO \times EE)^{(1/3)}$$

Where AO is Average Occupancy and EE is External Energy Efficiency

13.2.C Average Occupancy (AO)

Average Occupancy will be calculated as:

$$AO = PM / DD$$

Where PM is Person-Mile Distance and DD is Event Distance Driven

13.2.D Person-Mile Distance (PM)

Person-Mile Distance will be calculated for each Segment of the Tour or each Official FSGP Lap. Person-Mile Distance is the Event Distance Driven during a Tour Segment or FSGP lap multiplied by the minimum number of occupied seats within the solar car while driving that Tour Segment/FSGP lap. The summation of these Segment or Lap Person-Mile Distances will yield the total Person-Mile Distance, which is used in the scoring equation.

13.2.E External Energy Efficiency (EE)

External Energy Efficiency will be calculated as:

$$EE = DD / EU$$

Where DD is Event Distance Driven and EU is External Energy Usage

13.2.F External Energy Usage (EU)

Total External Energy usage of a Multi-Occupant vehicle will be calculated as:

$$EU = (n+1) \times Q + M$$

Where n is the number of times the energy storage system is charged from unmetered external energy between the start line and the finish line, Q is the energy capacity of the energy storage system in kWh, and M is the total metered external energy in kWh used to charge the energy storage system during the Event.

13.2.F.1 The energy capacity (Q) of the energy storage system will be calculated in kWh by the Event Organizers based on the manufacturer's battery cell datasheet and the number of cells in the system.

- 13.2.F.2** Multiple unmetered charging sessions will be counted as a single unmetered charge as long as the solar car is not driven in the Event between the charging sessions.
- 13.2.F.3** Charges from metered external energy shall be conducted in accordance with Reg 8.12. The energy from all metered external energy charges will be added up to calculate the total metered external energy in kWh. If for any reason an external charge is not properly metered and recorded, it will count as charging from unmetered external energy.

13.2.G Practicality Score (P)

- 13.2.G.1** Teams will be required to prepare a “window sticker” for their solar car which describes the attributes of the car. Teams will provide a digital copy of this “window sticker” to ASC Headquarters for publication through social media on the morning of judging.
- 13.2.G.2** The Practicality Score for a Multi-Occupant vehicle team will be determined by a panel of judges appointed by the ASC Organizers. It will be based on a mix of subjective and objective evaluations in the following areas:
- 1) Comfortable occupant seats that are easy to get in and out of
 - 2) Occupants are able to access a cup holder, sunglasses holder, and USB port while buckled into their seat
 - 3) Ergonomic steering wheel with easy to operate turn signals, horn, and other controls
 - 4) Clear driver visibility with wide field of view, good rearview camera view, and functional sun visors
 - 5) Easy to use vehicle infotainment system with good audio quality and bluetooth phone connectivity for music and navigation
 - 6) Spacious dedicated cargo area(s) separated from the occupant cabin, wheel wells, and vehicle electronics
 - 7) Occupant and cargo access doors that are well secured and easy to open/close from inside and outside
 - 8) Stylish interior with a good fit and finish
 - 9) Stylish exterior with a good fit and finish
 - 10) Overall vehicle desirability based on window sticker and in person vehicle evaluation
- 13.2.G.3** Practicality judging will be conducted on the solar cars during the rest day prior to the start of the Tour. The score from each judge will be averaged into the overall Practicality Score for the solar car ranging from 0-100%.
- 13.2.G.4** The MOV Practicality Judging Rubric will not be released to judges or teams until the morning of the judging.
- 13.2.G.5** Prior to the start of ASC, Officials will review and compile the Practicality Scores for all MOV teams and each team will be provided with their own score along with a copy of each judge’s MOV Practicality Judging Rubric evaluation of their vehicle. Practicality Scores of all teams will be made public at the ASC Awards Ceremony at the conclusion of the Event.

13.2.H ASC Stage Ranking

The scoring formula from Reg 13.2 is modified as follows for MOV stage ranking during the Tour:

$$S = 50 \times (D/D') + 50 \times (O/O')$$

The Practicality Score (P) is removed from the scoring equation. For the purposes of this calculation, External Energy Efficiency (EE) will be cumulative from the start of the Event. All other values will be from the single Stage.

13.2.I FSGP Ranking

The scoring formula from Reg 13.2 is modified as follows for MOV ranking in FSGP:

$$\text{Score} = 50 \times (D/D') + 50 \times (O/O')$$

The Practicality Score (P) is removed from the scoring equation for FSGP.

13.3 Solar Car Configuration

All practicality features shall remain intact throughout the duration of the Tour.

14 FSGP/Qualifier

14.1 Track Event Format

During ASC event years, the track event portion will serve as both a FSGP event and the Qualifier for the cross-country event (Tour) as outlined in Reg 12. Herein, this portion of the event will be named the "Track Event".

The Track Event consists of three (3) on-track days around a closed-course track.

14.2 Unit of Distance

The mile is the unit of distance for the Event.

14.3 Determination of Winner

14.3.A Single-Occupant Vehicles

Overall placing for the FSGP Event will be determined based on the team with the highest total number of Official FSGP Laps calculated by summation of the laps driven during Track Event hours minus penalty Deductions. If there is a tie for Official FSGP Laps, the team with the lowest individual lap time wins.

14.3.B Multi-Occupant Vehicles

Refer to Reg 13.2.I for ranking of Multi-Occupant vehicle teams.

14.4 Solar Car Configuration

Solar Cars must drive in the same configuration as approved during Scrutineering.

14.5 Radios / Communication

The team must be in two-way radio communication with the solar car driver at all times. Communications should be continuously maintained between the solar car, the pit area, and timing.

14.6 Authority

Headquarters reserves the right to cancel the track activity at any time for the Event as a whole or for any particular team.

14.7 Safety

14.7.A Safety Equipment

14.7.A.1 Teams are required to have the safety equipment readily available as outlined in Reg 3.1.B (with exception of the traffic cones and flags as these are only applicable for ASC).

14.7.A.2 Teams must have their battery spill kit available in the pit area at all times.

14.7.A.3 Teams shall have first aid supplies in their pit area at all times.

14.7.A.4 Any team member in the hot pit or going on the track to assist with a broken-down vehicle must be wearing a safety vest and closed toed shoes.

14.7.B Support Vehicles

No support vehicles will be allowed on the track without prior consent by Track Officials.

14.7.C Safety Vehicles

Designated safety vehicles will be driven by an Official(s). If a track safety vehicle is needed, it will attempt to stay in the slow lane of the track. The same rules that apply to passing solar cars also apply to passing the safety vehicle. Teams may pass a static safety vehicle during a yellow flag condition as well as any solar car not moving given the conditions are safe to pass.

14.8 Briefings

A Briefing will be held at the start of each Event day. A special meeting may be called in cases of emergency. Attendance at meetings by a team representative and driver(s) is required. Briefing notes and other daily updates will be available at Headquarters and/or communicated through each team's Generic Team Email. All official statements, rule interpretations, and special instructions will be contained in these postings.

14.9 Timing

14.9.A Responsibility

Headquarters will only recognize timing and distance information from the Timing Officials.

14.9.B Timers

Each team must provide a timing person. This team member must be in radio contact with both the solar car driver and the pit crew. The timer will be paired with a timer from another team, and the two will work together to ensure that every lap is properly recorded. Timing Officials will review all timing data for consistency and accuracy. The timer must report 15 minutes prior to the start of the Event. Any laps the solar car runs without a timer in place or without being recorded by the timer and verified by a partner team will not be counted.

14.9.C Official Time

A master clock providing the "Official Time" will be located near the starting line and timing area. A solar car must cross over the start line to begin and end an Official FSGP Lap. All timers must record lap start/finish times based on the Official Time displayed.

14.9.D Transponders

If available, the track's timing system will be utilized by Timing Officials for fast lap data and as a secondary reference for timing data if they have questions on a team's timing information. Teams shall be prepared to affix an electronic transponder to the solar car such that it reliably triggers the track's receiver (typically in a wheel fairing with a clear line of sight to the ground). Additional instruction will be provided at the Event.

14.9.E Track Hours

The track will be open for driving from 10:00 am – 6:00 pm local time (Day 1) and 9:00 am – 5:00 pm local time for Days 2 and 3.

14.10 Starts

14.10.A Group Start

Teams are released from the Start Line in a group start. The movement of all vehicles in the Start Line area will be under the control of the Start Line Officials.

14.10.B Starting Order

For teams that have yet to complete any FSPG laps, the starting order will be based on the same order that teams completed Scrutineering. Teams who have completed at least one valid FSGP lap will start ahead of those who haven't, with their starting order determined based on the total number of Official FSGP Laps completed thus far with ties going to the team with the lowest individual lap time.

14.10.C Start Line

Solar cars will be released simultaneously from the starting line at the beginning of each day. Teams must report to their starting position 15 minutes prior to the beginning of the start time. Control of solar cars in the start line area is under the direction of the Start Line Officials. Teams not ready or who are returning to the track from the cold pit area must wait for a Track Official to clear them for the starting line and be sequenced into traffic.

14.10.D Delayed Start

The start of the Track Event may be delayed if inclement weather or other hazardous conditions appear likely to pose a threat to the solar cars or their drivers. If the start of the Track Event is delayed, then the assigned start time for that day will be adjusted accordingly. The end of the day will not be adjusted.

14.11 End of Day

At 6:00 pm (Day 1) or 5:00 pm (Days 2 and 3) the Checkered Flag will be flown at the Start/Finish line. A team currently on a lap started in advance of the End of Day time will be given the opportunity to complete that lap, provided that the lap time for that final lap is no more than 20% greater than the time for the preceding lap.

14.12 ASC Qualification Requirements

14.12.A Vehicle Qualification

In order to qualify for the ASC Tour portion of the Event, each team shall obtain a minimum number of Official FSGP Laps. The minimum required total distance is:

- 200 mi (322 km) if achieved in one day
- Or 300 mi (483 km) if achieved in two consecutive days

14.12.B Driver Qualification

In addition, each driver must obtain a minimum number of laps driven. The minimum required total distance per driver is 50 mi (80 km). Laps still count towards driver qualification even if penalty laps have been applied against the team or laps have been forfeited (i.e. failing to impound overnight).

14.12.C Discretion

At the discretion of the Inspectors and the ASC Organizers, entrants who do not complete the requirements for Qualification may be permitted to participate in the Event. Additional requirements may be imposed on these entrants.

14.13 Charging/Impound

All battery enclosures shall be secured overnight in impound configuration by the Inspectors. Headquarters shall be appraised of special issues for impound.

14.13.A Off-track Charging

All solar cars may begin the Track Event with a fully charged battery pack. Once the Track Event begins, charging may only occur using the vehicle's solar collector.

14.13.A.1 A designated charging area will be provided for the morning and evening charging times, as well as for teams who wish to stop and charge in the middle of the day. Internal combustion generators will not be permitted within the charging area. Only solar car charging may occur within this designated area.

14.13.A.2 All charging of solar car batteries shall be monitored by a team member. Unattended charging of the solar car batteries is strictly prohibited.

14.13.A.3 Charging from other sources will result in a penalty associated with Official FSGP Laps completed to that point, reference Reg 15.5.J.

14.13.A.4 Multi-Occupant vehicles are allowed to re-charge their battery packs as per the procedure and scoring of Regs 13.2.F.3 and 8.11.

14.13.B Impounding

All registered and sealed batteries must be secured in the impound configuration (Reg 8.11) that was inspected during Scrutineering and will be secured by the Inspectors. Batteries must be impounded by 8:00 pm each evening and will be released from impound at 7:00 am the following morning.

14.13.B.1 If an SOV team doesn't impound overnight, they forfeit all Official FSGP Laps completed to that point. Reference Reg 15.5.M for failure to impound on time.

14.13.B.2 Multi-Occupant vehicles can maintain access to their vehicle for metered external energy charging. Battery impound is not required for MOV unmetered external energy charging. Any time a team elects to charge from unmetered external energy they must first officially declare this intention to the Event Organizers.

14.14 Driving Procedures

All solar cars will proceed around the course in the prescribed direction at all times. Passing and lane information will be presented at the pre-Track Event team meeting.

14.14.A Speed Limit

No solar car shall exceed the maximum speed of 65 mph (104.2 km/h), unless further restricted by the Officials per Reg 10.9.B anywhere on the track. All teams entering the pit area must slow down. A walking pace must be observed at all times in the cold pit area. While in the cold pit area a team member shall walk directly in front of the solar car to ensure that the path is clear and to warn others in the area.

14.14.B Following Too Closely

No solar car shall follow another vehicle too closely.

14.14.C Pushing

Except for the following situations, solar cars must not be pushed or pulled from the time they are moved into their starting position for the daily start until they reach the finish line except in the cold pit areas. In no case shall regenerative braking be engaged while pushing or pulling the solar car.

14.14.C.1 Garage Area: Solar cars may be pushed within the confined areas of the Cold Pit or Garage Area.

14.14.C.2 Emergency: In an emergency or breakdown situation, the solar car must be removed from the track as quickly as is prudent. In this circumstance, the car may be pushed or lifted off the track. Upon resuming driving, the solar car may then be pushed or lifted back onto the track to the same location where it left the track.

14.14.C.3 Penalty: If a solar car is pushed on the track, a penalty will be applied as per Reg 15.5.F.2.

14.15 Breakdowns

Any solar car that stops on the track because of mechanical problems, lack of power, or an accident must be moved off the track as soon as possible and prior to any repairs being performed.

All teams must carry a tow strap in their solar car at all times. This tow strap will be used to facilitate a tow to the pit area by the safety vehicle, if required. The tow strap must be sized appropriately to tow the solar car with a driver inside, must be long enough to extend at least 3 m (10 ft) in front of the leading edge of the vehicle when installed, and must have closed hooks on both ends. No ratchet straps are allowed. Laps where a solar car is towed in are invalidated and do not count as laps driven.

Team members must not cross the track to their solar car unless permission is granted by a Track Official. All team members attending to a broken-down car on the track must be wearing a safety vest.

At the discretion of the Inspectors, any solar car that has a breakdown/accident, or undergoes alterations/repairs for any reason, may be required to be re-inspected for safety and rules compliance.

14.16 Team Uniforms

For the duration of the Event, team members shall wear uniforms representing their host Institution(s) from 7:00 am to 8:00 pm. The uniforms are required to have the Institution name, team number, and Event logo. Artwork for the Event logo can be downloaded from the Event Website.

14.17 Track Operation Flags

Flags will be used to provide trackside instruction to solar car drivers.

- **Master Flag Position:** The Master Flag Position will be near the starting line and display flag(s) to represent the overall condition of the track. Corner workers will be located at various Flag Positions around the course to display “local” flag conditions.
- **Green Flag:** Track clear; proceed at your chosen speed. This flag will be displayed at the starting line only.
- **Blue Flag:** Caution ahead, proceed at your chosen speed, passing is allowed. This flag will be displayed at corners to alert drivers to unusual conditions ahead that do not immediately interfere with the track (such as a car stopped off of the track).
- **Yellow Flag:** Caution, obstruction on or near track. A yellow flag means that cars are to slow down and no passing is allowed unless waved around by a Track Official. The no passing rule will continue to apply until the solar car reaches a Flag Position where no flag is displayed. Whenever a yellow flag is somewhere on the track, the Master Flag Position will display both the green and yellow flags.
- **Black Flag:** Return to the pit area immediately. A black flag is given to an individual car at which point the driver must return to the pit area. The black flag will appear at the last corner before pit entry.
- **Red Flag:** Total stoppage due to major accident or some other reason. All cars must pull to the side and stop where they are on the track without passing. Proceed only when instructed to do so by Track Officials.
- **White Flag:** This flag will be displayed at the starting line only at five (5) minutes before the End of Day time and signifies that there is less than five (5) minutes remaining in the race day.
- **Checkered Flag:** This flag will be displayed at the starting line only at the End of Day time and signifies that the race day is over, save and except for the stipulation in Reg 14.11.

15 Penalties

Any team failing to comply with these Regulations during Scrutineering, FSGP, or ASC will be penalized. Penalties range from official warnings to disqualification from the Event. It is the responsibility of the Chief Inspector and/or Track Steward, with input from the other Inspectors and the Observers and track personnel, to determine whether an infraction occurred, the severity of the incident, and the appropriate penalty. All penalties will be submitted by the Chief Inspector and/or Track Steward to Headquarters for subsequent posting. Disqualification of a team from the Event requires concurrence of the Event Director. For the Tour portion of the Event penalty Deductions will be applied to Official Distance. For the Track Event penalty Deductions will be applied to the total number of Official FSGP Laps.

15.1 Penalty Distance/Laps

15.1.A Penalty Distance

All penalty distances listed are suggested minimums. For each Tour infraction, Penalty Distance calculations will be rounded up to the next mile. Driving conduct penalties may double with each subsequent infraction. Scrutineering and other technical penalties will normally be the same for each infraction. If the Inspectors believe the teams are deliberately violating traffic or driving regulations for strategic advantage, they may impose more severe penalties up to and including potential disqualification.

15.1.B Penalty Laps (Track Event)

For each Track Event infraction, Penalty Distance will be converted to Penalty Laps based upon the length of the track (rounded up to the next lap). Penalty Laps are deducted from laps driven to determine Official FSGP Laps.

15.1.C Scrutineering Issues

Scrutineering issues may result in a daily loss of laps (Track Event), and loss of distance (Tour) as determined by the Inspectors. Examples of Scrutineering issues include but are not limited to: car size exceeds limits. Typically, these issues are the result of receiving a Blue status in any Scrutineering inspection area. Issues within the dynamics area may also result in a reduced speed limit for the solar car on the track or on-road.

15.2 Posting of Penalties

Teams will be notified of the details of their penalties via their Generic Team Email.

15.2.A Tour: Penalties will be publicly posted the evening before the start of the next Stage. On the last day of the Tour, penalties per team will be posted no later than 2 hours after the finish of the Tour or the arrival of a team's car, whichever is later.

15.2.B Track Event: Penalties will be publicly posted each evening. On the last day of the Track Event, penalties will be posted no later than 2 hours after the finish of the Track Event.

15.3 Protests

Protests may be filed for any reason, including disputing a penalty levied against any team, correcting timing errors in the Official published results, or protesting the actions of another team. Protests must be submitted by email to Protest@AmericanSolarChallenge.org. A "filing fee" Deduction of 5 miles will be assessed against the team's Official Distance or Official FSGP Laps for the day on which the protest is filed. The Jury will hear all protests. Jury meetings will be held in private. A representative of the team(s) involved may attend deliberations concerning disputes regarding their team to give evidence and respond to questions from the jury, but not the vote. A representative of the Event may attend deliberations to give evidence and respond to questions from the jury, but not the vote.

15.3.A Opportunity to Be Heard

Protests will normally be heard by the Jury at the earliest possible Jury sitting. It may be necessary in some instances for the Jury to postpone the hearing on a protest.

15.3.B Time Limit

15.3.B.1 Tour: Except for the last day, all protests against penalties must be filed by 9:00 am the morning after the penalty is posted. Protests that do not directly relate to a penalty must be filed by 6:30 pm on the last day of the Stage when the issue occurred. On the last day of the Tour, protests for any purpose must be filed within 30 minutes after the finish of the Tour, after a team arrives, or after a team is notified of their penalties, whichever is later.

15.3.B.2 Track Event: Except for the last day, all protests against penalties must be filed by 9:00 am the morning after the incident occurred, or after the team is notified of the penalty they are disputing, whichever is later. On the last day of the Track Event, protests must be filed within 30 minutes after the finish of the Track Event or after a team is notified of their penalties, whichever is later.

15.3.C Protest Judgements

The decision of the Jury is final and no further appeals are allowed. The Jury will notify Headquarters of their decision, and the Chief Inspector and/or Track Steward or delegate will then inform the affected teams. The Jury may refund some or all of the filing fee, which will be credited to the day the filing fee was assessed.

15.4 Conduct

Penalties, including disqualification from the Event, may be imposed for improper conduct or the use of alcohol or illegal substances. Improper conduct may include, but is not limited to, improper language, unsportsmanlike conduct, unsafe behavior, or cheating. Teams are responsible for the conduct of all persons associated with the team, whether or not they are officially registered.

15.5 Penalties

The following outlines some predetermined penalty distance values.

15.5.A Speeding

Any solar car found to be speeding will be penalized. Speeding penalties may be assessed based on the following factors: (1) velocity over posted speed limits, (2) length of time of speeding infraction, (3) location of speeding infraction (i.e. work zones, school zones, etc. where typical penalties would be doubled automatically). The speed of either the solar car itself or the chase vehicle may be used in determining a speeding infraction. Speeding infractions including a rate of speed of 70 mph (112.6 km/h) or greater will result in a 10 mile penalty in addition to the calculated assessment.

15.5.B Traffic Violations

- 15.5.B.1 Traffic Violations:** Any solar car committing a traffic violation may be penalized, up to disqualification. Any solar car driver who commits three (3) traffic violations (including speeding) over the course of the Tour may be individually disqualified from the Event.
- 15.5.B.2 Traffic Violations by Support Vehicles:** If a support vehicle is observed to be committing a traffic violation may result in a penalty applied to the team.
- 15.5.B.3 Track Operation Violations:** Any solar car committing a track operation violation may be penalized, up to disqualification. Any solar car driver who commits three (3) track violations (including speeding) over the course of the Track Event may be individually disqualified from the Event.

15.5.C Failure to Allow Other Traffic or Solar Car Caravans to Pass

Any solar car caravan that fails to properly facilitate passing by traffic or other solar car caravans may be penalized a minimum of 5 miles.

15.5.D Support Vehicles Impeding Tour

Any support vehicles too close to solar car caravans or impeding another team's solar car caravan will be penalized a minimum of 5 miles.

15.5.E Drafting

A minimum 2 mile penalty may be assessed for any time a solar car drafts behind another vehicle.

15.5.F Pushing

- 15.5.F.1** A 1 mile penalty for every 30 seconds a team pushes may be assessed each time a team pushes or pulls their solar car in order to advance along the Tour Route. (Except in the exclusions as per Reg 12.26)
- 15.5.F.2** A loss of the current lap will result each time a team pushes or pulls their solar car in order to advance along the track. (Except in the exclusions as per Reg 14.14.C)

15.5.G Improper Ballast

A 25 mile penalty may be assessed each time a team operates their solar car with ballast that does not match the solar car occupant.

15.5.H Unauthorized Drivers / Occupants

- 15.5.H.1** Any solar car that is driven with an unauthorized driver or contains an unauthorized occupant will be required to return to the starting point of the infraction and drive with an authorized personnel in order to receive credit for driving beyond that point.
- 15.5.H.2** Any solar car that is driven on the track with an unauthorized driver or contains an unauthorized occupant will forfeit double the number of laps driven with this personnel.

15.5.I Driving Outside of Authorized Times

Teams will be penalized two (2) miles for every one (1) minute they continue to drive past the time they were supposed to stop.

Teams will be penalized two (2) miles for every one (1) minute they start driving earlier than the time they were supposed to start.

15.5.J Non-Solar Charging of Batteries

After the start of the Track Event / Tour until the Official finish, teams will be disqualified from the Event for charging their solar car's storage batteries from any source other than those allowed by Reg 8, without specific written instruction from the Officials. Such charging of a solar car storage battery will constitute replacement and is subject to Reg 15.5.L.

Multi-Occupant vehicles are exempt from this penalty.

15.5.K Disturbing Official Battery Seals

Solar car batteries will be marked with an official seal. Disturbing these seals in a manner that prevents proper identification by Inspectors may be penalized as though all of the battery modules affected had been replaced as in Reg 15.5.L.

15.5.L Replacement of Batteries

Decisions to exchange (or externally recharge – see Reg 15.5.J) all or part of a battery must be communicated formally to the team's Observer or an Inspector. The penalty will be calculated as:

$$\text{Distance penalty (miles)} = 480 \times (n+S) / N$$

where:

n = number of replacement modules

S = sum of all modules previously replaced

N = total number of modules in solar car battery pack

15.5.M Failure to Impound

Teams will be penalized two (2) miles for every one (1) minute they fail to Impound their batteries.

15.5.N Exceeding Size Specifications

Oversized solar car bodies will be penalized 7 miles per day per excess of 0.1m² of planview area (length x width).

Oversized solar car bodies will be penalized 7 miles per day per excess of 0.1m² of planview area

If both the car body and solar collector are oversized, both penalties will be applied. Each penalty will be pro-rated if the oversize area for body or solar collector is less than 0.1m², with a minimum of 2 miles per day for oversized solar collectors, and a minimum of 1 mile per day for oversize solar cars

15.5.O Securing of Fasteners

Failure to comply with Reg 10.4.B Securing of Fasteners will result in a penalty of 1 mile per bolt per day per instance where proper securing is not applied. The Head Mechanical Inspector shall determine which non-compliant fasteners must be brought into compliance with Reg 10.4.

15.5.P Parking Brake Penalty

A 10 mile per day penalty will be applied for a non-functioning parking brake based on Reg 10.6.

15.5.Q Roll Cage Clearance Penalty

A 2 mile per day penalty will be applied for each centimeter of clearance lacking between the roll cage padding and a driver's helmet based on Reg 10.3.A.14. It will be at the discretion of the Inspectors whether less clearance will be allowed with application of the penalty.

15.5.R Ignoring BPS Fault Indicator

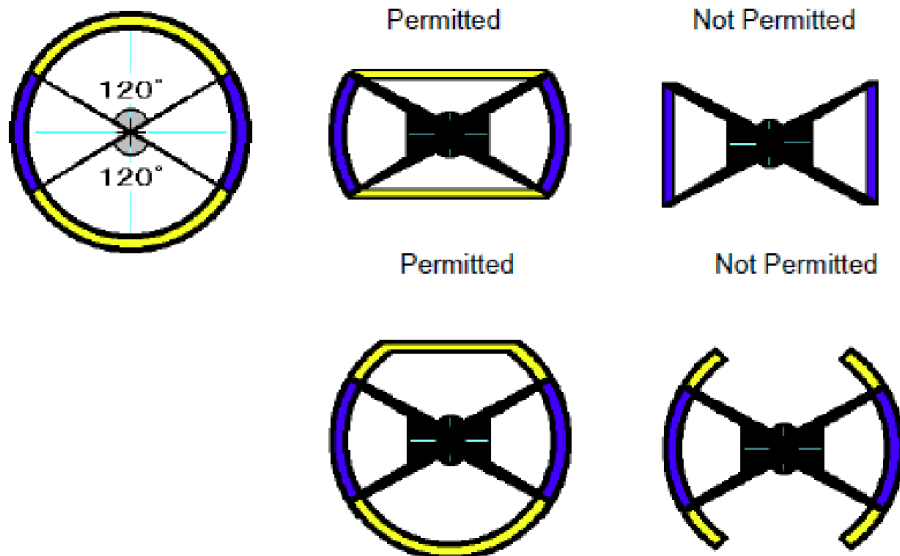
A 2 mile penalty will be applied for each occurrence that the team ignores the BPS Fault Indicator and continues to drive along the route.

Appendix A Steering Wheel Specifications

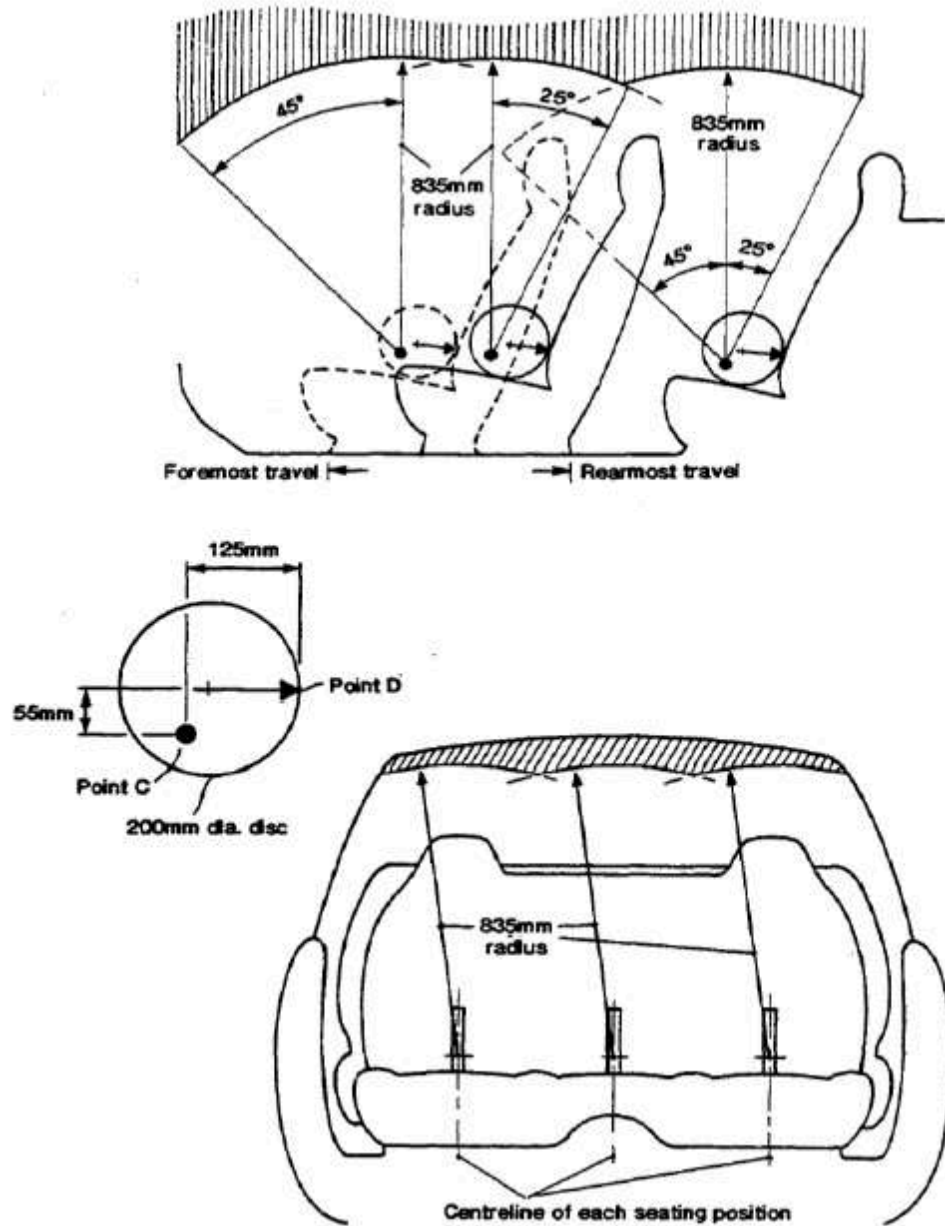
© 2006/08 with acknowledgement to Japanese Automobile Federation

To reduce the possibilities of driver injury in the event of collision and to minimize impediments to emergency egress, the steering system must be controlled by a steering wheel which has a continuous perimeter.

A circular shape is preferred, however the upper part above 2/3 and/or the lower part below 2/3 of the circumference of the steering wheel may be flat as depicted in the diagram below).



Appendix B Occupant Space Diagram



Appendix C Reference Standard for Lighting

Reference Reg 9.4.

The reference standard lights are:

- Red, Amber and White: Item MS6 TruFLEX Auxiliary Light Strips from Custom Dynamics
 - 6 LED (Length 2.3 in)
 - <https://www.customdynamics.com/magic-spots-6-led-light-bar>

The reference standard lights will be used in the following manner:

- Lights shall be powered by a regulated 12 volt power source.
- Light shall be centered on a flat surface with a matte white finish.
- Reference standard lights shall be placed next to the solar car's lights to compare during inspections.
Only one color of reference light shall be illuminated at a time.

Appendix D Mechanical PVDR Form

School Name _____ Team Number _____

Mechanical PVDR/Table of Contents

1. History of vehicle (one paragraph) page _____
2. **Type of vehicle:** Single-Occupant (___), Multi-Occupant (___) check one
3. Complete vehicle mass in driving configuration: with batteries, with ballasted driver/occupants, etc. (estimate) (____); Units (___) kg (___) lbs
4. Vehicle description shall be presented by profile and top view drawings showing the configuration and number of wheels, placement of major components such as driver, battery, ballast box, seat belts mounting points, etc, along with overall dimensions including wheel base and track width page _____
5. **Frame/chassis and roll cage type:** tubular frame (___), composite (___), check one. Drawing shall show the (1) occupants positioned in the frame/chassis, (2) material specs of all metal components, and (3) compliance with Reg 10.3.A. page _____
6. **Roll cage:** Profile and frontal drawings shall include material specs and show compliance with Regs 10.3.A, 10.3.B, 10.3.C page _____
7. **Seat Belts:** 5 point (___), 6 point (___), check one page _____
Drawing shall indicate location of mounting points and compliance with Reg 10.3.E
8. **Braking system:** Front wheel only (___), Front-rear (___), check one. page _____
Schematic and description of primary braking system shall include parking brake and component specs demonstrating compliance with Regs 10.5 and 10.6 (include brake calcs for force to stop in time and force to lock the front wheels)
9. **Steering system type:** rack and pinion (___), other (___), check one. page _____
Description shall include component selection and specs
10. **Front suspension type:** a-arm (___), other (___), check one page _____
Description shall include drawing/photos, component specs, and engineering analysis demonstrating proper selection and sizing of rod ends with shear loads under applied loads as specified in Appendix E, section E.2
11. **Rear Suspension Type:** a-arm (___), swing arm (___), other (___), check one. page _____
Description shall include drawing/photos, component specs, and engineering analysis demonstrating proper selection and sizing of rod ends with shear loads under applied loads as specified in Appendix E, section E.2
12. Description of method of analysis to be used to prove structural integrity page _____
For loading conditions as per Appendix E, Section E.2
13. **Windshield:** material (_____) page _____
Include data sheet showing compliance with Reg 9.7.B.2

Mechanical contact: Name: _____
Email address: _____
Phone: _____

Project Manager: Name: _____
Email address: _____

Appendix E Mechanical Report Instructions

It is the intent of the Event to maintain the safest entries possible while encouraging a wide variety of designs and materials for solar-powered vehicles. The purpose of the mechanical report is to require each team to document, by calculation or testing, the structural integrity of their vehicle and the protection provided to their drivers. Note that each team is responsible for the safety of its members, and any minimum criteria specified by the Event shall not be construed as design specifications for the construction of a "safe" solar vehicle.

The mechanical report shall be presented in the format described in this document. Reports that do not meet this format will not be read and could jeopardize a team's chances for competing in the Event. The report should be written to be as concise as possible and should only present information relevant to the requirements of the report. The report is limited to a total of thirty (30) pages (not sheets) in length, including appendices.

E.1 Report Presentation

Teams shall completely fill out the Mechanical VDR Form (Appendix F) and include it as the cover page and table of contents at the beginning of the report.

E.2 Loading Conditions

Teams shall consider road and traffic conditions when determining loading criteria and potential failure modes. For suspension and steering systems, analysis shall include a minimum of **1G turn, a 2G bump, and 1G braking** case loads with a worst-case condition of the loads combined. These loads shall to be applied at the **tire contact patch**. Loads shall be provided in dimensional units, along with any relevant assumptions used in the analysis.

E.3 Vehicle Impact Analysis

The vehicle impact analysis section must include the following topics:

E.3.1 Specifications: The report must describe the vehicle frame and construction techniques (aluminum space frame, composite monocoque, etc.), including the materials utilized, their important dimensions (e.g., tubing diameters and thicknesses, number and types of plies in composite constructions), and their properties (in the "as welded" or "as fabricated" condition). The report must also list the specific impact criteria that are assumed for each case, as well as sample calculations and computer output as applicable. Other relevant assumptions used in the analysis should be listed.

E.3.2 Drawings: The report must include structural drawings of the vehicle from five viewpoints: top, front, side, rear, and isometric. These drawings must illustrate the following:

- Driver location and orientation
- All members considered "structural"
- Locations of ballast and batteries
- Locations of chassis hard points (points of attachment).
- Calculated center of mass

The report must contain structural drawings of the driver's compartment from three viewpoints: top, front, and side. These drawings must illustrate the following:

- Driver location
- Roll cage design and location
- Location of structural members
- Driver's harness attachment points

The report must contain an isometric drawing of the body and solar collector, as well as show the egress opening latch design. All drawings must be identified by number and must include a description.

E.3.3 Analysis: Analyses may be in the form of computer modeling (such as a finite-element analysis) or empirical testing of the actual vehicle or its components. For finite-element analysis of the roll cage, 3D elements shall be used for analysis of all joints. Either a full 3D element model of the roll cage shall be used, or a shell or beam model of the roll cage shall be used to set the boundary conditions for detailed 3D models of each joint that extend at least 2x the tube diameter from the toe of the weld. Shell or beam elements are acceptable for analysis of the remainder of the Occupant Cell. Shell elements shall be acceptable for finite-element analysis of the roll cage if the factor of safety is 1.4 or greater. In any case where a minimum factor of safety of less than 1.1 is reported via finite-element analysis, additional data shall be required to validate the finite element model.

E.3.3.1 Occupant Cell Impact: Front, rear, and side impact with another vehicle assumes a bumper height of 100 mm, a width of 600mm, and elevation off the ground of 350 mm as shown in Figure 5. The Occupant Cell shall not deform more than 25 mm and shall not exceed ultimate strength, but can yield (Reg 10.3.A.9). The required load cases are:

1. Front
2. Rear
3. 3 side impact locations. One of these locations shall be the worst case bending moment on the side of the Occupant Cell.

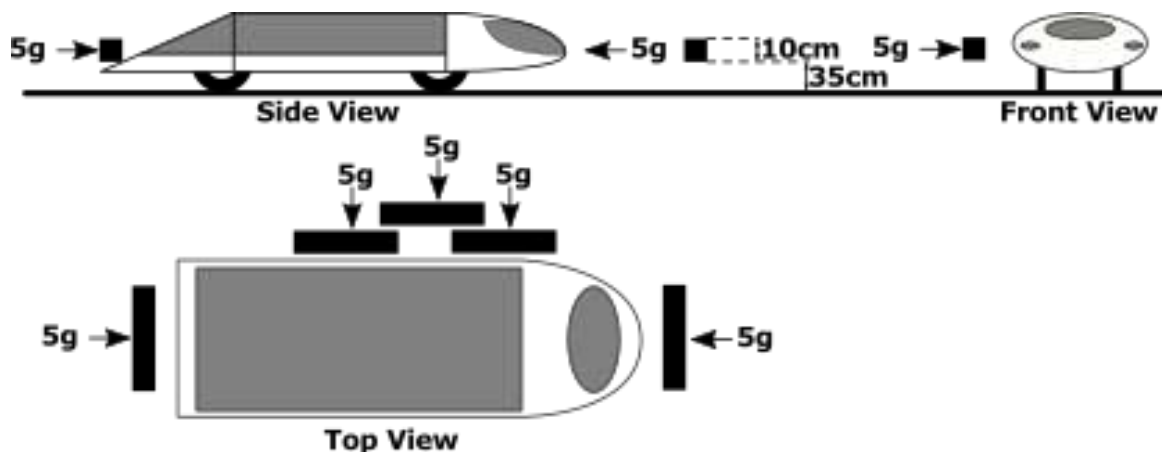


Figure 5 Occupant Cell Impact Loads

E.3.3.2 Roll Cage Impact: Roll cage impact scenarios shall have a loading patch no more than 150mm diameter. The roll cage can not exceed yield strength (Reg 10.3.A.9). The required load cases are:

1. Combined Loading (5g down, 4g backward, 1.5g lateral)
2. Sideways angled loading (5g at 30 degrees downward from horizontal)
3. Sideways angled loading (5g at 60 degrees downward from horizontal)
4. Sideways horizontal loading (5g at the top of the hoop)
5. Rearward horizontal loading (5g at the top of the hoop)

Each load case should be evaluated when applied to the front and rear roll hoop (10 load cases total). An example of the combined roll cage loading case for the front hoop is included in Figure 6.

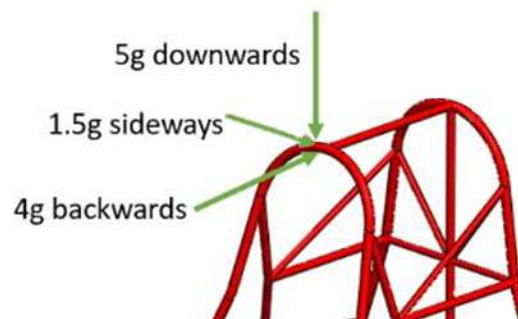


Figure 6 Combined Roll Cage Load Case (Front Hoop)

Emphasis should be placed on how protection is provided for the driver under these conditions. All impact scenarios must take into account movement of body panels and the vehicle's solar collector to ensure that these members do not penetrate the space occupied by the driver during the impact.

E.3.4 Conclusions: The report should contain a summary of the findings for each impact case and a brief discussion of the results. Raw data should be attached as an appendix to the report.

E.4 Vehicle Stability Analysis

Address vehicle stability with respect to Reg 10.2.C, Consider vehicle center of gravity and relative weights on each wheel. Documentation with calculations and/or testing should be provided. Photos, drawings and anecdotal references are acceptable.

E.5 Design Review

Attach a completed design review form from the Team Status Board.

E.6 Appendix

The appendix should only include relevant information that is referenced from the main body of the report. Appendix items should be organized into two appendices. Examples of what could possibly be included in the appendices are: detailed calculations and results, empirical testing data, details on construction techniques, and component specifications that are particularly relevant to the analysis.

- **Appendix A** - Mechanical Systems Analysis Supporting Documentation
- **Appendix B** - Vehicle Impact Analysis Supporting Documentation

Appendix F Mechanical VDR Form

School Name _____ Team Number _____

Mechanical VDR Form/Table of Contents

1. History of vehicle (one paragraph) page _____
2. **Type of vehicle:** Single-Occupant (___), Multi-Occupant (___) check one
3. Complete vehicle mass in driving configuration: with batteries, with ballasted driver/occupants, etc. (estimate) (____); Units (___) kg (___) lbs
4. Vehicle mass distribution (estimate), front (____), rear (____), lbs/kg
5. Vehicle static stability analysis as specified in Appendix E, section E.4 page _____
6. Vehicle description shall be presented by profile and top view drawings showing the configuration and number of wheels, placement of major components such as driver, battery, ballast box, seat belts mounting points, etc, along with overall dimensions including wheel base and track width page _____
7. **Frame/chassis and roll cage type:** tubular frame (___), composite (___), check one. Drawing shall show the (1) occupants positioned in the frame/chassis, (2) material specs of all metal components, and (3) compliance with Reg 10.3.A page _____
8. **Roll cage:** Profile and frontal drawings shall include material specs and show compliance with Regs 10.3.A, 10.3.B, 10.3.C page _____
9. **Seat Belts:** 5 point (___), 6 point (___), check one page _____
Drawing shall indicate location of mounting points and compliance with Reg 10.3.E
10. **Braking system:** Front wheel only (___), Front-rear (___), check one. page _____
Schematic and description of primary braking system shall include parking brake and component specs demonstrating compliance with Regs 10.5 and 10.6 (include brake calcs for force to stop in time and force to lock the front wheels)
11. **Steering system type:** rack and pinion (___), other (___), check one. page _____
Description shall include component selection and specs
12. **Steering stops:** Description/drawing/photos shall show compliance with Reg 10.7.B page _____
13. **Front suspension type:** a-arm (___), other (___), check one page _____
Description shall include drawing/photos, component specs, and engineering analysis demonstrating proper selection and sizing of rod ends with shear loads under applied loads as specified in Appendix E, section E.2
14. **Rear Suspension Type:** a-arm (___), swing arm (___), other (___), check one. page _____
Description shall include drawing/photos, component specs, and engineering analysis demonstrating proper selection and sizing of rod ends with shear loads under applied loads as specified in Appendix E, section E.2
15. **Tires and rims:** Description shall include brand, load, speed, and pressure rating to comply with Reg 10.2 page _____
16. **Hub design:** Assembly cross section view of king pin/hub/brake disc/bearings/spindle page _____
17. **Battery enclosure:** Description/drawing to show how battery enclosure is constructed and secured in the chassis as per Reg 8.4.B page _____
18. Description/drawing of independent solar collector attachment methods per Reg 10.1.C page _____
19. **Fasteners:** Description of compliance with Reg 10.4 page _____
20. **Vehicle Impact Analysis:** Method: Classical (___), FEA (___), Testing (___) page _____
Analysis shall be performed as per Appendix E Section E.3 and the results shall be presented in terms of factor of safety in tabulated form

Mechanical contact: Name: _____
Email address: _____
Phone: _____

Project Manager: Name: _____
Email address: _____

Appendix G Battery Approval Form



Battery Approval Form

The manufacturer's specification sheet, the battery's MSDS sheet with accident protocol, and a description of the protection circuitry (protection circuitry schematic, high level description, list of items protected) must also be submitted to Headquarters prior to approval. Battery approval is subject to verification at Scrutineering. If the manufacturer changes the battery's specifications, the new specifications must be submitted for re-approval. Teams must bring a minimum of 3 unmodified cells for Scrutineering verification.

Note: For this form, the term "battery" refers to the smallest single unit produced by the manufacturer (usually containing one cell but could be a module or the entire pack in the case of a commercially available supplemental battery). Teams or suppliers may group batteries together to form "modules". The term "pack" refers to the full vehicle battery system made up of multiple batteries.

TEAM CONTACT INFORMATION

School Name: _____	Date: _____
Team Number: _____	Team Battery Contact Email: _____
Team Email: _____	Team Battery Contact Phone #: _____

BATTERY MANUFACTURER'S TECHNICAL SPECIFICATIONS

Manufacturer URL: _____	Type/Chemistry: _____
Battery Name: _____	Model Number: _____
Capacity (Ah): _____	Voltage (Min/Nom/Max): _____
Mass (kg): _____	Max Charge (A or C): _____
Cost (US\$): _____	Max Discharge(A or C): _____

BATTERY PACK SPECIFICATIONS

# of batteries in pack: _____	Mass (kg): _____
Capacity (kWh): _____	Voltage (Min/Nom/Max): _____

Pack Configuration (cell connections & # of cells per module, modules per string, strings in parallel, etc): _____

BATTERY SUPPLIER INFORMATION

Information for the company that is supplying the batteries to the team. This may be the original manufacturer or a reseller. If the supplier uses a different model name or number than the manufacturer, please provide that information.

Supplier: _____	Contact: _____
Email: _____	Phone: _____
Battery Name: _____	Supplier Model #: _____

Appendix H Recommendations

These are not binding parts of the Regulations, as they are impractical to enforce fairly, but the Officials believe they are essential to creating a good, safe solar car. All teams should meet these recommendations on their own.

Driver Training: Driver Training is essential to ensuring your car can be driven safely. Solar vehicles have substantially different controls, field of vision, and feedback from regular vehicles. It is recommended that every team trains all drivers in a gradual and controlled manner. This means that a driver will begin learning to handle the vehicle in a very controlled setting (such as a large closed parking lot or track) and progress to more challenging situations as he/she gains experience. Prior to any driving, the vehicle should always be given a safety check and the driver should have properly functioning safety equipment. As training progresses, the driver (and all team members in the caravan) should practice handling the following situations:

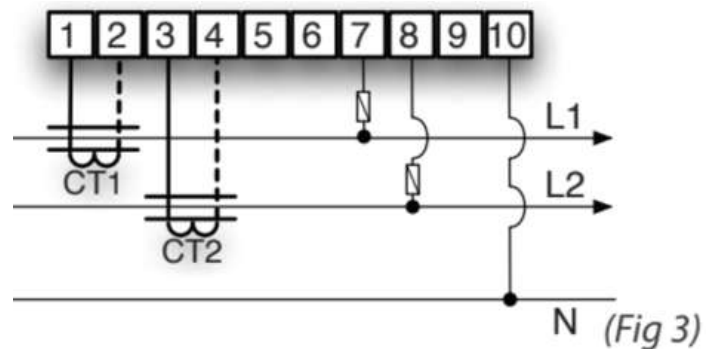
- (1) Blow-outs and run flats, especially of the single rear wheel on a three-wheel vehicle
- (2) Loss of power
- (3) Primary brake partial and complete failure
- (4) Passing protocols
- (5) Merging protocols
- (6) Road hazard avoidance
- (7) Moving to shoulder from operational speed
- (8) Caravan communication and protocols for possible emergencies
- (9) Emergency stop and egress

Vehicle Stability and CG: CG location has an important influence on vehicle stability. Your CG location should be a design requirement, not something you find after you build the car. Information about recommended CG location, and its influence, will be posted on the Event Website.

Wheel Base / Width Ratio: It is recommended that the wheel base to width ratio be greater than 1.5.

Appendix I Onboard MOV Energy Meter Overview

- This onboard energy meter will be provided to MOV teams by IEF
- The meter must be installed in the solar car during Scrutineering
- EKM Omnimeter Pulse v.4 kWh meter provides revenue grade metering accuracy
- Flexible 120-240 Vac input voltage at up to 40A current
- Watertight enclosure with sealed cable glands
- ~\$500 estimated cost for this charge meter solution

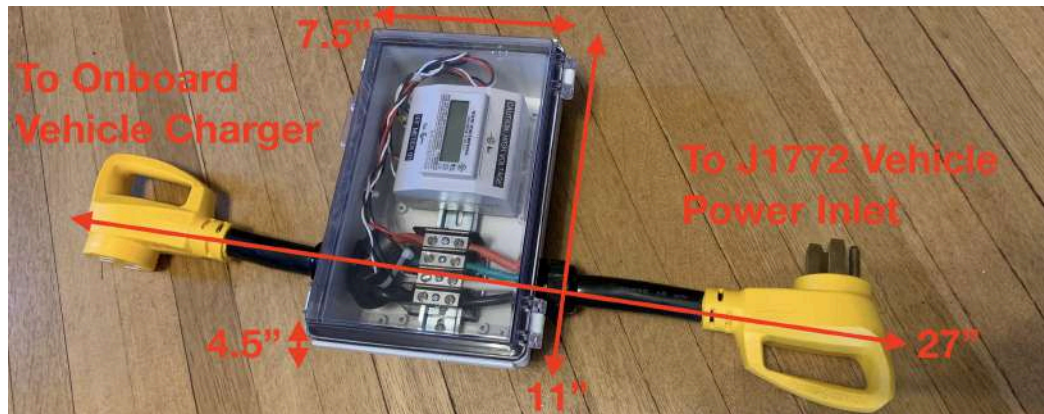


Meter BOM (for Reference)

Item	QTY	Unit Cost	Total Cost
EKM Omnimeter Pulse v.4 – Universal Smart Meter	1	\$ 260.00	\$ 260.00
EKM Split-core CT, 200A Current Transformer	2	\$ 45.00	\$ 90.00
EKM Watertight Enclosure with Hinged and Latching Lid (6.7" x 10.63" x 4.33")	1	\$ 80.00	\$ 80.00
Camco 55215 18" PowerGrip Extender - 50 AMP	2	\$ 19.95	\$ 39.90
Marathon Special Products 1414300 Splicer Block, 4-P, 115A, Line: 2AWG-14AWG	1	\$ 18.08	\$ 18.08
Waterproof Cable Gland (2pc Set)	1	\$ 10.05	\$ 10.05
Total			\$ 498.03

Meter Packaging

The kWh hour meter is 6lbs, watertight, and has enclosure dimensions of 11" L x 7.5" W x 4.5" H with a connector to connector length of 27".



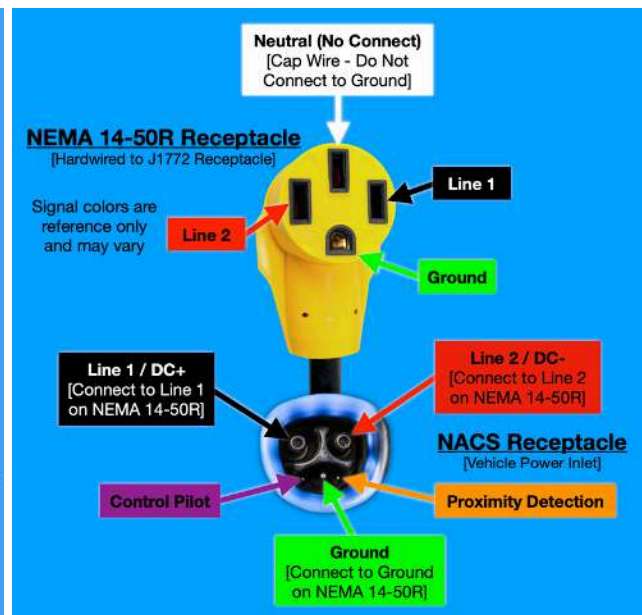
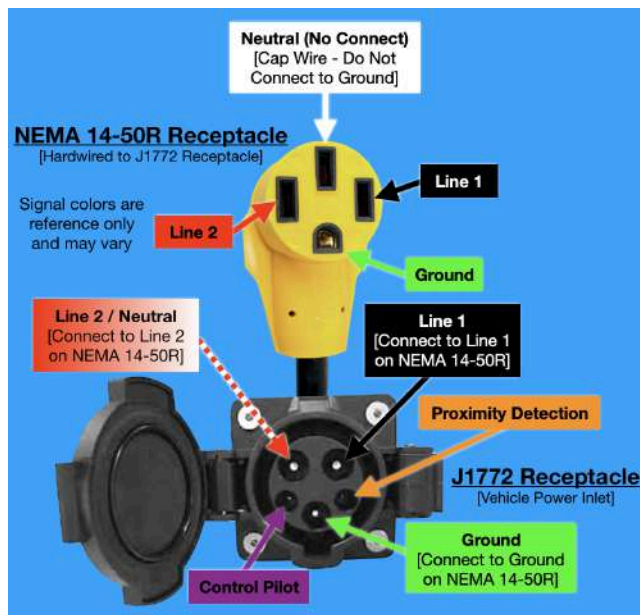
Solar Car Integration

MOV solar cars must feature an inline NEMA 14-50 connection between the J1772 or NACS Vehicle Power Inlet and the Onboard Vehicle Charger. When not at an ASC/FSGP Event, the NEMA 14-50 connectors will be directly connected as shown below.



When the team comes to an ASC/FSGP Event, they will disconnect the NEMA 14-50 connectors and plug them into the kWh Charge Meter as shown below. The NEMA connectors will be sealed together for the duration of the Event so the meter can't be bypassed.

The J1772 or NACS receptacle and NEMA 14-50 receptacle shall be wired as follows to match the meter:



Appendix J Official Regs Interpretations

These interpretations are current as of the publishing date of these regs revision, see the website for any new updates:

<https://www.americansolarchallenge.org/american-solar-challenge/2026-event/registration-and-regulations-2026/#interpretations>

- Reg 8.7.A.4 (Issued May 8, 2026):

In the off position, the Main Power Contactors must electrically isolate the high voltage battery, motor, and solar collector from each other and put the solar car in the 'Safe State'. The high voltage battery pack contactors must be electrically located first in series after the battery terminals, or second in series after the main fuse.

The portion of interest specifically concerns the suitability of multiple, parallel main battery pack contactors. The event has had a long standing unofficial interpretation of not allowing parallel contactors. The official request has afforded us the opportunity to revisit this interpretation, and after careful deliberation we've decided to allow multiple, parallel main battery contactors, as long as they collectively conform to the regulations as written in 8.7.A. This may be good news for several teams, as the previous interpretation has been a fairly common point of contention in regards to pre-charge topology. As a corollary, we'll also allow multiple main pack fuses as long as they collectively conform to 8.6.A, but only if they're used to protect different circuits rather than to increase capacity to a single circuit.

For teams that received push-back during design review and already made changes that are no longer necessary, you're welcome to make adjustments given the new interpretation, and invited to re-submit updated VDRs as you wish. If you're already green, then it's up to you whether you want to re-submit or proceed at risk until pre-event scrutineering.

- Reg 10.3.A.14 (Issued May 20, 2026):

The headrest may wrap around the back half of the occupant's head. The headrest is not called out as part of the occupant cell in 10.3.A.6, nor does 10.3.A.15 mention the headrest as an object that requires 30mm/50mm of clearance. However, a headrest that wraps around the back of the helmet still must not restrict the occupants's head movement to the point that it impairs their ability to look around.

Appendix K Revision Table

Rev	Description	Date
A	- Significant cleanup/re-arrangement of regs throughout the document - PVDR/VDR instruction previously split between Reg 5 and various appendices; redundant/conflicting info. Eliminated all but the Mechanical VDR instructions appendix and otherwise centralized all instructions in Reg 5. - Cleanup of some language in Reg 6 - Reg 7: Changed from "Grandfathered" to "Demonstration" class, re-wrote Reg 7.1.C. Cars designed for future events can compete in Demo class (eg, bring your fall WSC car to test at FSGP in the summer) - Reg 8.1.B: SOVs solar collector area bumped up to 6sqm - Reg 8.1.H: No more water spraying arrays - Reg 8.2.A: Batteries now sized based on capacity, not mass. - Reg 8.4.B: Added G loading considerations for battery enclosure mounting. - Reg 8.4.D: Clarified battery ventilation requirements - Reg 8.5: New reg clarifying acceptable cooling - numerous shenanigans over the years with teams asking if they can pack ice around their motor controller or in their battery intake, carrying the battery into a convenience store refrigerator in the middle of the day, etc. - Reg 8.7.A: Language change from "main switches" to "main contactors" - Reg 8.12: MOVs can charge off NACS as well as J1772 - Reg 9.1: Allowed vehicle dimensions updated to larger WSC 2025 size - Reg 9.4: Eliminated DRLs - Reg 9.5: Moved horn into its own heading instead of under "Lighting" - Reg 9.6: Moved "outside air circulation" under its own heading instead of under "Visibility" - Reg 9.7.D: Re-wrote forward and side vision section, complete with diagram of inspection points. - Reg 9.8.B: Eliminated egress opening outline requirement - Reg 10.3.A.12: Added 15 degree requirement instead of just "angled backwards" - Reg 10.3.B.8: Added reg on seat movement/adjustability - Reg 10.5.E: Left foot brake pedals specifically allowed - Reg 10.7.C: Turning radius reduced from 16m to 15m - Reg 10.9.B: MPH as the primary speed unit (also everywhere after) - Reg 10.9.E: New acceleration test (proxy for hill climb) - Moved "Team Uniforms" into the FSGP section and all other sections about uniforms reference it, rather than duplicating language. - Significant cleanup of language throughout Reg 12 - Complete rewrite of MOV scoring in Reg 13 - Significant cleanup of language throughout Reg 14	December 18, 2024

B	• 2.10: Confirmed that the naming sponsor will be Electrek • Several places: Updated language from “Team Data Sheet” to “Team Roster” • 3.1.B.2: Changes that the safety kit SHOULD travel with the solar car to SHALL travel with the solar car • Reg 4.1.A.1 through 4.2.B, and 4.5: replace TBDs with actual dates and fees • 4.4: Re-wrote requirements for the Team Data (Vehicle Data Sheet, Team Photo, and Team Roster) • 5.3.B.4: clarified language • 5.4.B.2: Clarified that this applies for both the main and supplemental pack • 5.3.B: Clarified some wording about what we’re looking for in the electrical VDR. Re-ordered, with the impound stuff all in the bottom three bullets • 5.3.B.2, C.1, and D.2: clarified “Main and supplemental pack” rather than “each battery type” • 5.3.C.4: New • Numerous places: standardized the language as “battery enclosure”, not “battery box” • 5.3.E.2 3): require receipts • 5.3.E.2 9) and 8.1.E: Only three (3) different types or sizes of solar cells, not 6 • 8.1.A: Allowing Silicon-pervoskite tandem solar cells • 8.2: only need to supply sample cells for the main battery. Clarified language about how energy capacity is determined • 8.2.A: clarifying that these capacity limits are for the main battery • 8.2.C: cleanup to match the language from FSGP 2025 regs. • 8.2.C.4: removed • 8.2.D: updated language • 8.3: the protection circuitry SHALL be designed to allow for testing, and SHALL be contained in the battery enclosure(s) • 8.3.A.5: removed “with vehicle not in motion”. We have never inspected or actually required BPS systems to be unable to clear faults with the vehicle in motion. • 8.3.B.1: Clarified that this only applies to the main battery • 8.3.B.2: Allow lithium supplemental batts, IF they are commercially available with an integrated/packaged BMS • 8.4.A: Updated isolation requirements • 8.4.D: Updated language on when the battery ventilation must be running • 8.4.E: Removed, redundant with 8.5 • 8.5: Updated language • 8.6.A: Updated language. Fuse need not be on the high side of the pack, mid-pack or low-side fuses are acceptable • 8.6.C: Cell voltage taps may not leave the battery enclosures. • 8.7.A.3: Removed prescriptive labeling requirements. • 8.7.A.4: Re-worded given the changes in 8.6.A about the fuse • 8.7.C.1: Clarified location of external power cut switch • 8.10: updated “Low voltage” to be 60V, in line with NEC standards	November 19, 2025
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	• 8.11.B.3: Language update • 9.3: Dropped ground clearance requirement to 75mm • 9.4: Updated that the light width measurements are to the outer edge of the light. • 9.4.A, 9.4.B, and 9.4.C: eliminated the maximum distances from the front and rear of the car: if we can see the lights from the required angles, they're fine. • 9.4.D: clarified that the high-mount center brake light has a distance below the highest *fixed* point of the car. • 9.4.E.2: Clarified language. Set a time that the fault light must activate within. • 9.7.B.4: Should -> Shall • 9.8.A: Eliminated the secondary egress direction requirement for SOVs. Clarified that the egress test starts with the parking brake off. • 9.8.B.4: Updated egress marking requirements • 9.9.C.3: clarified language • 9.9.E: Other ballast types are acceptable at the discretion of the event officials • 9.10.A.3: Solid color brush stroke • 9.10.E.3: Should -> Shall • 10.2.A.2: updated language • 10.3.D: Clarified language about the requirements. • 10.3.E.5: seatbelts with certs that expire mid-event are not allowed • 10.4.A: Should -> Shall • 10.4.B (3): safety wire may not be used in sheer. Nordlock washers are explicitly disallowed as a fastener securing method. • 10.4.C: language cleanup • 10.4.E.1: Should -> Shall • 10.4.F: clarified language • 10.5.G.3: language cleanup • 10.7.C: eliminated clearance requirement; U-turn is to the contact patches. Matches WSC requirement. • 10.9.E: Added in acceleration time. • 11.2.B: Updated to disallow SNELL 2010 (will be >12y old before the event) and allowed SNELL 2025, ECE 22.06, and FAI helmets. Added notes about disallowing "DOT 218" helmets that clearly don't actually meet DOT requirements. • 11.4: added that the driver must be able to drink with at least one hand on the steering wheel • 11.5.A: Cell phones are not allowed to be the primary comms method with the solar car, period. 12.6.B and language in 14.5 about cell phone eliminated as redundant. • 12.7.C.1: Updated that the team may designate the passenger seat in the front row as the observers seat • 12.14.A: Checkpoint hold time typical value dropped from 45 to 30 minutes • 12.16.A: Updated typical checkpoint and loop hold times from 45->30 minutes and 15->5 minutes, respectively. • 12.17.A.2: Should -> Shall • 14.9.D: Updated language	
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	• 14.13: Should -> Shall • 14.13.A: Clarified language. Re-ordered sub-sections • 15.3.B.2: Clarified language. • 15.5.N: clarified language • Appendix D, E, F: cleanup of forms	
C	• New 4.1.A.2: Other team registration items will not be processed before team entry form • 4.1.C.3: Fixed regs# reference • 4.2.A.7: added date • 5.1: Date in file name of submissions • 5.2.A.2: include steady state load transfer • 9.3: clarified ground clearance regs. we will not be enforcing a speed limit on ground clearance configs • 10.4.B (1): officially added Fujilok to the list of approved fasteners • 12.16: added 12.16.B about checkpoints with multiple loop options. • 12.22: teams must also observe minimum posted speed limits • Appendix C: officially changed the reference light to the MS6 light for red, amber, and white • Added new Appendix J with official regs interpretations that have been posted to the website.	May 25, 2026